MATH 205

GROUPS

DEF: Let G be a non-empty set together with an operation $*$ on G, i.e. $*$ is a function from $G \times G$ to G.

$(* : G \times G \to G)$

However, rather than writing $\times(g,h)$ we write $g \times h$

Such that

(i) $*$ is associative, i.e. that for every $a, b, c \in G$.

$$a*(b*c) = (a*b)*c$$

(ii) There is an identity element e for the operation, i.e. that

$$\forall g \in G, \quad e*g = g = g*e \quad (1)$$

NOTE If an identity exists then it is unique.

Suppose that $\exists e' \in G$ s.t.

$$e'*g = g = g*e', \ \forall g \in G. \quad (1')$$

$$\underset{\substack{(1') \\ w/\ g=e}}{e} = e*e' \underset{\substack{(1) \\ w/\ g=e'}}{=} e'$$

(iii) Every element in G has an inverse, i.e

$\forall g \in G$ there exist a $g' \in G$

s.t $g*g' = e = g'*g$

NOTE. If an inverse exists then it is unique.

Suppose that there exists a g'' s.t.

$$g*g'' = e = g''*g \quad (2')$$

Claim $g' = g''$

Proof

$$g'*(g*g'') \underset{(2')}{=} g'*e = g'$$

$\|$ associativity

$$(g'*g)*g''$$

$\|$ (2)

$$g'' = e*g''$$

ABEL

DEF: Suppose that $(G, *)$ is a group

We say that $*$ is commutation (or G is abelian) if

$$\forall a, b \in G, \ a*b = b*a$$

NOTATION We sometimes denote the group operation by multiplication, i.e. We write $g \cdot h$ instead of $g \times h$, if no confusion should arise.

In this case we denote the identity by 1, and the inverse of g by g^{-1}

If G is abelian, it is more common to denote the group operation by $+$.

In this case, we denote the identity by 0 and the inverse of an element $g \in G$ by $-g$.

$\mathbb{N} = \{0, 1, 2, \ldots\}$ natural

$\mathbb{Z} = \{\ldots -2, -1, 0, 1, 2, \ldots\}$

$\mathbb{Q} := \{\frac{p}{q} \mid p, q \in \mathbb{Z}, q \neq 0\}$ rational numbers

$\mathbb{R}$ = real numbers

$\mathbb{C}$ = complex numbers $= \{a + ib \mid a, b \in \mathbb{R}\}$, $i^2 = -1$

EXAMPLES

(i) $(\mathbb{N}, +)$
- associative ✓
- identity ✓
- inverse ✗

not a group

(ii) $(\mathbb{Z}, +)$
$(\mathbb{Q}, +)$
$(\mathbb{R}, +)$
$(\mathbb{C}, +)$

group

(iii) $(\mathbb{Z}, \cdot)$

associative ✓
identity ✓
inverse ✗

not a group

(iv) $(\mathbb{Q}, \cdot)$

assoc ✓
ident ✓
inverse ✗

not a group

(v) $(\mathbb{Q}\setminus\{0\}, \cdot)$

group

(vi) $\mathbb{Z}_n = \{0, 1, \dots, n-1\}$, $n \geq 1$

$\mathbb{Z}_n$ has two operations $+$ and $\cdot$.

$(\mathbb{Z}_n, +)$ is also an abelian group.

(vii) Let $S_n = \{f \mid f: \{1, \dots, n\} \to \{1, \dots, n\}\}$ s.t. f is a bijection.

$|S_n| = n!$

There is a natural operation on S_n, defined as

$$(f \circ g)(x) = f(g(x)) \text{ for every } x \in \{1, \dots, n\}$$

OBSERVATION

$(S_n, \circ)$ is a group.

- $(f \circ g) \circ h = f \circ (g \circ h)$ assoc ✓
- $i(x) = x$, $\forall x \in \{1, \dots, n\}$
- $f \circ i = f = i \circ f$ identity ✓
- $f \circ f^{-1} = i = f^{-1} \circ f$ inverse ✓

Note that if $n \geq 3$ then $(S_n, \circ)$ is not abelian.

$$\begin{pmatrix} 1 & 2 & 3 & 4 & \cdots & n \\ 3 & 2 & 1 & 4 & \cdots & n \end{pmatrix} \circ \begin{pmatrix} 1 & 2 & 3 & 4 & \cdots & n \\ 2 & 1 & 3 & 4 & \cdots & n \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 2 & 3 & 4 & \cdots & n \\ 2 & 3 & 1 & 4 & \cdots & n \end{pmatrix}$$

$$\overset{?}{\neq} \begin{pmatrix} 1 & 2 & 3 & 4 & \cdots & n \\ 2 & 1 & 3 & 4 & \cdots & n \end{pmatrix} \circ \begin{pmatrix} 1 & 2 & 3 & 4 & \cdots & n \\ 3 & 2 & 1 & 4 & \cdots & n \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 2 & 3 & 4 & \cdots & n \\ 3 & 1 & 2 & 4 & \cdots & n \end{pmatrix}$$

TERMINOLOGY

Suppose $f: X \to Y$

- f is one-to-one $\Leftrightarrow$ f is injective
- f is onto $\Leftrightarrow$ f is surjective

bijective

NOTATION

if $f \in S_n$ we write

$$\begin{pmatrix} 1 & 2 & \cdots & n \\ f(1) & f(2) & \cdots & f(n) \end{pmatrix}$$

instead of

$f \in S_3$ $f(1) = 3$
$f(2) = 1$
$f(3) = 2$

$$f \leftrightarrow \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}$$

G always denotes a group.

Suppose that $H \subseteq G$

$$\begin{array}{ccc} H \times H & \dashrightarrow & H \\ \cap\!| & & \cap\!| \\ G \times G & \longrightarrow & G \\ (x,y) & \longrightarrow & xy \end{array}$$

Suppose that if $x, y \in H$ then $xy \in H$, $\forall x, y \in H$

Then we can ask whether H with this restricted operation is a group.

In order for this to be true, it has to have on identity, Call it e_H

Then $e_H \cdot e_H = e_H$ in G.

$e_H^{-1}(e_H e_H) = e_H^{-1} \cdot e_H = e_G$ MULTIPLY

$= (e_H^{-1} e_H) e_H = e_G \cdot e_H = e_H = e_G$

If H is a group w/ the induced operation then

$$1 = e_G \in H$$

Moreover for any $h \in H$, it has to have an inverse i.e

$$\exists h^* \in H \text{ s.t } h^* h = 1 = h \cdot h^*$$
$$\Rightarrow h^* = h^{-1} \in H$$

Hence if it is a group w/ induced operation, the following properties have to be satisfied.

(i) $1 \in H$

(ii) $\forall h_1, h_2 \in H \quad h_1 h_2 \in H$

(iii) $\forall h \in H$, $h^{-1} \in H$

DEF: If $H \subseteq G$ s.t. (i)(ii)(iii) are satisfied then we say that H is a subgroup of $G \rightarrow H \leq G$

<u>Claim</u>

Let $H \subseteq G$. Then

$H \leq G \Leftrightarrow$ (i) $1 \in H$

(ii) $\forall x, y \in H$
$xy^{-1} \in H$

▶ $(\mathbb{Z}, +)$ Then let $n \in \mathbb{N}$

$n\mathbb{Z} := \{na \mid a \in \mathbb{Z}\}$

then $n\mathbb{Z} \leq \mathbb{Z}$

$2\mathbb{Z} = \{\ldots, -4, -2, 0, 2, \ldots\}$

$O := \{a \in \mathbb{Z} \mid a \text{ is odd}\}$
$= \{\ldots, -5, -3, \ldots\} \subseteq \mathbb{Z}$

not a subgroup.

COSETS OF A SUBGROUP

Let $H \leq G$.

Then define the following equivalence relations on G.

(i) $x \sim_H y \Leftrightarrow y^{-1}x \in H$

(ii) $x \sim_H y \Leftrightarrow yx^{-1} \in H$

<u>Recall</u>

Associated to E, we have for every $s \in S$

$[s] := \{t \in S \mid (s,t) \in E\}$
equivalence class of s

$\bigcup_{s \in S} [s] = S$

$[s] = [t] \Leftrightarrow (s,t) \in E$

$\forall s, t \in S$ there are two possibilities.

either $(s,t) \in E$ in which case $[s] = [t]$

OR $(s,t) \notin E \quad [s] \cap [t] = \emptyset$

If $s_i \in S$, w/ $i \in I$ are s.t.

- $(s_i, s_j) \notin E$ for $i \neq s$
- And for every $s \in S$, $\exists i \in I$ s.t $(s, s_i) \in E$

then $S = \bigcup_{i \in I} [s_i]$

(i) $\sim_H$: Let $x \in G$

$$[x] = \{y \in G \mid x \sim_H y\}$$
$$= \{y \in G \mid y^{-1}x \in H\}$$
$$= \{y \in G \mid x^{-1}y \in H\}$$
$$= \{y \in G \mid y \in xH\}$$
$$= xH$$

inverse

"The left coset of H in G which contains x"

$\mathcal{L}_H(G) := \{xH \mid x \in G\}$

→ the set of left cosets of H in G.

$$\bigcup_{xH \in \mathcal{L}_H(G)} xH = G \qquad |G| = \sum_{xH \in \mathcal{L}_H(G)} |xH| = \sum_{xH \in \mathcal{L}_H(G)} |H|$$

WE USE THAT CLAIM

$$= |H| \cdot |\mathcal{L}_H(G)|$$

Claim

$$|xH| = |H|$$

Proof

$$x^t : H \longrightarrow xH$$
$$h \longrightarrow xh$$

x^t is bijective so $|xH| = |H|$ □

DEF: If $|G| < \infty$ then $|G|$ is called the order of G.

- $|\mathbb{Z}_n| = |\{0, 1, \dots n-1\}| = n$
- $|S_n| = n!$

If $|G| < \infty$, we say that G is a finite group.

$\mathbb{Z}$ is not a finite group.

Proposition $\boxed{\text{(Lagrange)}}$

If $|G| < \infty$, and $H \leq G$ then

$|H| \mid |G|$ ($\mid$ divides)

[**DEF** well defined $\in$ or $\notin$ clear input = output]

Proof

$$|G| = |H| \cdot |\mathcal{L}_H(G)|$$
$$\Rightarrow |H| \mid |G|$$

□

(ii) $\sim_H$: Let $x \in G$

$$[x] = \{y \in G \mid x \sim_H y\}$$
$$= \{y \in G \mid yx^{-1} \in H\}$$
$$= Hx$$

right coset of H in G which contains x.

$$\mathcal{R}_H(G) = \{Hx \mid x \in G\}$$
$$G = \bigcup_{Hx \in \mathcal{R}_H(G)} Hx$$

$$|Hx| = |H|$$
$$\Rightarrow |G| = |H| \, |\mathcal{R}_H(G)|$$

Claim $|\mathcal{L}_H(G)| = |\mathcal{R}_H(G)|$

PROOF

$$\alpha : \mathcal{L}_H(G) \longrightarrow \mathcal{R}_H(G)$$
$$\alpha : xH \longrightarrow Hx^{-1}$$

α is well-defined.

If $xH = x'H$, we need to show $Hx^{-1} = H(x')^{-1}$

def of α

Since $xH = x'H$; $x' = xh$

Hx^{-1} and $H(x')^{-1} = H(xh)^{-1}$
$$= Hh^{-1}x^{-1}$$
$$= Hx^{-1}$$

Check: well-defined

⚠ $\beta : \mathcal{R}_H(G) \longrightarrow \mathcal{L}_H(G)$
$$Hx \longrightarrow x^{-1}H$$

$$\beta \circ \alpha = id = \alpha \circ \beta$$

□

In general

$$\mathcal{L}_H(G) \neq \mathcal{R}_H(G)$$

(If abelian for example, they are equal.)

▶ $G = S_3 = \{ \underset{I}{\begin{pmatrix}1&2&3\\1&2&3\end{pmatrix}} \underset{\sigma}{\begin{pmatrix}1&2&3\\2&3&1\end{pmatrix}} \underset{\sigma^2}{\begin{pmatrix}1&2&3\\3&1&2\end{pmatrix}}$ $\quad \tau_i^2 = I$

$\underset{\tau_3}{\begin{pmatrix}1&2&3\\2&1&3\end{pmatrix}} \underset{\tau_1}{\begin{pmatrix}1&2&3\\1&3&2\end{pmatrix}} \underset{\tau_2}{\begin{pmatrix}1&2&3\\3&2&1\end{pmatrix}} \}$

$H = \{I, \tau_3\}$, $H \leq G$

$\mathcal{L}_H(G) = \{H, \tau_1 H, \tau_2 H\}$

$\{\tau_2, \tau_2\tau_3\} = \{\tau_2, \sigma\}$

$\{\tau_1, \tau_1\tau_3\} = \{\tau_1, \sigma^2\}$

All elements of G

$\mathcal{R}_H(G) = \{H, H\tau_1, H\tau_2\}$

$\{\tau_2, \tau_3\tau_2\} = \{\tau_2, \sigma^2\}$

$\{\tau_1, \tau_3\tau_1\} = \{\tau_1, \sigma\}$

$\mathcal{L}_H(G) \neq \mathcal{R}_H(G)$

"Left coset which contains σ^2 is not the right coset which contains σ^2"

HW $G = S_3$ $H = \{1, \sigma, \sigma^2\}$ $H \leq G$ $\mathcal{L}_H(G) = \mathcal{R}_H(G)$

$\Rightarrow \mathcal{L}_H(G) = \{H, \tau_1 H\}$.

$\Rightarrow \mathcal{R}_H(G) =$

DEF: Let $H \leq G$ then we say that H is normal in G, if $\mathcal{L}_H(G) = \mathcal{R}_H(G)$. In this case we write $H \trianglelefteq G$.

If G is abelian and $H \leq G$ then

$H \trianglelefteq G$ (b/c $\forall g \in G$ $gH = Hg$)

Proposition Let $H \leq G$ then $H \trianglelefteq G \Leftrightarrow$ for every $g \in G$, $g^{-1}Hg = H$

(NOTATION for any subgroup $H \leq G$ Let $H^g = g^{-1}Hg$) $H^g \leq G$.

PROOF

($\Rightarrow$) Assume $H \leq G$. Hence

$\mathcal{L}_H(G) = \mathcal{R}_H(G)$

Let $g \in G$. $Hg \in \mathcal{R}_H(G) = \mathcal{L}_H(G)$

$\Rightarrow Hg = gH$

$\Rightarrow g^{-1}Hg = H$

($\Leftarrow$) Suppose that $g^{-1}Hg = H$, $\forall g \in G$

WANT $\mathcal{L}_H(G) = \mathcal{R}_H(G)$

PROOF

Let $gH \in \mathcal{L}_H(G)$

$g^{-1}Hg = H \Rightarrow Hg = gH$

$\Rightarrow gH \in \mathcal{R}_H(G)$

$\mathcal{L}_H(G) \subseteq \mathcal{R}_H(G)$

similarly $\mathcal{R}_H(G) \subseteq \mathcal{L}_H(G)$

Hence $\mathcal{L}_H(G) = \mathcal{R}_H(G)$

$H \trianglelefteq G$

Observation

Suppose $H \leq G$ and we want to define a natural group operation on $\mathcal{L}_H(G)$ (Respectively $\mathcal{R}_H(G)$)

The operation on $\mathcal{L}_H(G)$ (respectively $\mathcal{R}_H(G)$) would have the property,

$$(g_1 H)(g_2 H) := g_1 g_2 H$$

$$(\text{resp. } (Hg_1)(Hg_2) = H g_1 g_2)$$

For this operation to be well-defined, we need the following condition if $g_1 H = g_1' H$ and $g_2 H = g_2' H$ then

$$g_1 g_2 H = g_1' g_2' H$$

(Resp. if $Hg_1 = Hg_1'$ and $Hg_2 = Hg_2'$ then

$$Hg_1 g_2 = Hg_1' g_2'$$

THEOREM The above definition for the operation on $\mathcal{L}_H(G)$ (respectively for $\mathcal{R}_H(G)$) is well-defined if and only if

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$$H \trianglelefteq G$$

PROOF

($\Leftarrow$) On $\mathcal{L}_H(G)$ the operation is well defined means $\forall g_1, g_2, g_1', g_2' \in G$

$(g_1 H = g_1' H$ and $g_2 H = g_2' H \Rightarrow g_1 g_2 H = g_1' g_2' H)$

$\Leftrightarrow (g_1^{-1} g_1' \in H$ and $g_2^{-1} g_2' \in H \Rightarrow (g_1 g_2)^{-1} g_1' g_2' \in H)$

$\Leftrightarrow (g_1^{-1} g_1' \in H$ and $\underbrace{g_2^{-1} g_2'}_{h_3} \in H \Rightarrow g_2^{-1} \underbrace{g_1^{-1} g_1'}_{h_1 \in H} g_2' \in H)$

(Condition is satisfied)

first side If $H \trianglelefteq G$

$$g_2^{-1} \cdot h_1 \cdot g_2'$$

$$= \underbrace{g_2^{-1} \cdot g_2'}_{h_3 \in H} \underbrace{(g_2')^{-1} h_1 \cdot g_2'}_{h_2 \in H g_2' = H} \in H$$

since $H \leq G$

since $H \trianglelefteq G$

So if $H \trianglelefteq G$ then the oper. on $\mathcal{L}_H(G)$ is well-defined.

($\Rightarrow$) Conversely suppose that the operation on $\mathcal{L}_H(G)$ is well-defined.

WANT $H \trianglelefteq G$

PROOF

Need to show that $\forall g \in G$, $g^{-1} H g = H$

Take $g^{-1} h g \in g^{-1} H g$.

$$g^{-1} h g \in [\underbrace{(g^{-1} H)}_{g^{-1}h} \underbrace{(g H)}_{g} = e H = H]$$

$\Rightarrow g^{-1} H g \subseteq H$ **(1)**, $\forall g \in G$

Replace g with g^{-1}.

$g H g^{-1} \trianglelefteq H \Rightarrow H \subseteq g^{-1} H g$ **(2)**

(1) and (2) $\Rightarrow g^{-1} H g = H$, $\forall g \in G$

$\Rightarrow H \trianglelefteq G$

(Analogous proof for $\mathcal{R}_H(G)$)

Observation

Suppose that $H \trianglelefteq G$, then

$$\mathcal{L}_H(G) = \mathcal{R}_H(G) = G/H$$

and

$$\begin{array}{ccc} (G/H) \times (G/H) & \longrightarrow & G/H \\ \downarrow & & \\ (g_1/H, g_2/H) & \longrightarrow & g_1 g_2/H \\ \| & & \| \\ (Hg_1, Hg_2) & \longrightarrow & Hg_1 Hg_2 \end{array}$$

is well-defined.

Claim With this operation G/H is a group.

PROOF

(i) $(g_1H\, g_2H)\, g_3H = g_1g_2H\, g_3h = (g_1g_2)g_3H$

$= g_1(g_2g_3)H = g_1Hg_2g_3H = g_1H(g_2Hg_3H)$

↳ G is assoc.

(ii) $(gH)(eH) = geH = gH = egH = (eH)(gH)$

$\Rightarrow eH = H$ is the identity.

(iii) $(gH)(g^{-1}H) = gg^{-1}H = eH = g^{-1}gH$
$= (g^{-1}H)(gH)$

$\Rightarrow (gH)^{-1} = g^{-1}H$.

▶ Let $n \geq 1$,

$n\mathbb{Z} := \{nz \mid z \in \mathbb{Z}\}$

$\trianglelefteq \mathbb{Z}$

▶ $0+5k,\ 1+5k, \dots, 4+5k$

$\mathbb{Z}/n\mathbb{Z} = \{0+n\mathbb{Z},\ 1+n\mathbb{Z}, \dots (n-1)+n\mathbb{Z}\}$

↳ arithmetic modulo n

HOMOMORPHISM / ISOMORPHISM

$$\mathbb{Z}/2\mathbb{Z} = \{0+2\mathbb{Z}, 1+2\mathbb{Z}\}$$

	$0+2\mathbb{Z}$	$1+2\mathbb{Z}$
$0+2\mathbb{Z}$	$0+2\mathbb{Z}$	$1+2\mathbb{Z}$
$1+2\mathbb{Z}$	$1+2\mathbb{Z}$	$0+2\mathbb{Z}$

Suppose that G and G' are two groups

Let $\phi : G \to G'$ be a bijection s.t.

$\forall g_1, g_2 \in G$

$$\phi(g_1 g_2) = \phi(g_1) \cdot \phi(g_2)$$

Then ϕ is isomorphism.

G and G' are isomorphic.

OBSERVATION

Being isomorphic is an equivalence relation on the set of all groups.

We say that $\phi : G \to G'$ is a homomorphism if

$$\phi(g_1 g_2) = \phi(g_1)\phi(g_2) \quad \forall g_1, g_2 \in G$$

ANALOGOUS

$L : V_1 \to V_2$ linear trans.

$L(v+w) = L(v) + L(w)$

$L(\lambda . v) = \lambda L(v)$

13.12 THEO PG 128 **THEOREM**

Suppose $\phi : G \to G'$ be a homomorphism

(i) $\phi(1_G) = 1_{G'}$

$\phi(1) = \phi(1.1) = \phi(1)\phi(1)$

$1_G = \phi(1)^{-1}\phi(1) = \phi(1)^{-1}\phi(1)\phi(1)$
$= \phi(1_G)$

(ii) $\phi(g^{-1}) = \phi(g)^{-1}$

$1 = \phi(1) = \phi(g g^{-1}) = \phi(g)\phi(g^{-1})$

$\phi(g)^{-1}(\phi(g) \cdot \phi(g^{-1})) = 1$

$= 1 \cdot \phi(g^{-1}) = \phi(g^{-1})$

(iii) $\operatorname{im}(\phi) \leq G'$

$1_{G'} = \phi(1_G) \in \operatorname{im}(\phi)$

Let $x, y \in \operatorname{im}(\phi)$

$x = \phi(\alpha)$ $y = \phi(\beta)$

$xy^{-1} = \phi(\alpha)\phi(\beta)^{-1} = \phi(\alpha)\phi(\beta^{-1})$

$= \phi(\alpha\beta^{-1}) \in \operatorname{im}\phi$

(iv) $\ker\phi := \{g \in G \mid \phi(g) = 1\}$ 13.15 THEO PG 137 13.20 ALSO.

<u>Claim</u> $\ker\phi \trianglelefteq G$

<u>Proof</u> $1 \in \ker\phi \Leftrightarrow \phi(1) = 1$

$x, y \in \ker\phi$, $\phi(x) = 1$, $\phi(y) = 1$

$\Rightarrow \phi(y^{-1}) = \phi(y)^{-1} = 1$

$\phi(xy^{-1}) = \phi(x)\phi(y^{-1}) = 1.1$

$\Rightarrow xy^{-1} \in \ker\phi$

$\ker\phi \leq G$.

In order to show $\ker\phi \trianglelefteq G$ we need to show that

$g^{-1}\ker\phi\, g \subseteq \ker\phi \quad \forall g \in G$

Let $a \in \ker\phi$ want to show

$g^{-1}ag \in \ker\phi$

$\Leftrightarrow \phi(g^{-1}ag) = 1$

$\Leftrightarrow \phi(g)^{-1}\underbrace{\phi(a)}_{1}\phi(g)$

$= \phi(g)^{-1}\phi(g) = 1$

$N \trianglelefteq G$

$\Leftrightarrow \forall g \in G$ $g^{-1}Ng = N$

$\Leftrightarrow \forall g \in G$ $g^{-1}Ng \leq N$

g yerine g^{-1} alınca

$gNg^{-1} \leq N$

multiply with g^{-1}

$N \leq g^{-1}Ng$

Recall

Let $H \leq G$, we would like to define an operation on $\mathcal{L}_H(G)$ (resp. $\mathcal{R}_H(G)$)

s.t $(g_1H)(g_2H) = g_1g_2H$

(resp: $(Hg_1)(Hg_2) = Hg_1g_2$)

This operation is in general <u>not</u> well-defined. We proved that this operation on $\mathcal{L}_H(G)$ (resp on $\mathcal{R}_H(G)$) is well-defined $\Leftrightarrow H \trianglelefteq G$

And in this case, we showed that <u>this operation</u> makes

$G/H := L_H(G) = R_H(G)$, <u>a group</u>. This group is called the quotient (or factor group) of G by H.

-Let G and G' be two groups, we defined what it means for $\phi: G \longrightarrow G'$ to be a homomorphism, i.e. $\forall x, y \in G$ $\phi(x \cdot y) = \phi(x)\phi(y)$

-We proved

$\{g \in G \mid \phi(g) = 1\} =: \underline{\ker\phi \trianglelefteq G}$

Observation 1

Let $\phi: G \longrightarrow G'$ be a homomorphism.

<u>CLAIM</u> ϕ is injective $\Leftrightarrow \ker\phi = \{1\}$

<u>PROOF</u>

($\Rightarrow$) Suppose that ϕ is injective and $x \in \ker\phi$. Then

$\phi(x) = 1_{G'} = \phi(1_G)$

$\Rightarrow \phi(x) = \phi(1) \underset{\phi,\, inj}{\Rightarrow} x = 1$

So $1 \in \ker\phi \subseteq \{1\} \Rightarrow \ker\phi = \{1\}$

since $\ker\phi \leq G$

($\Leftarrow$) Suppose $\ker\phi = \{1\}$

Suppose that $\phi(x) = \phi(y)$

$\Rightarrow \phi(xy^{-1}) = \phi(x)\phi(y)^{-1} = 1 \Rightarrow xy^{-1} \in \ker\phi = \{1\}$

$\Rightarrow x$

(ii) Let $\phi : G \longrightarrow G'$ be a homomorphism

<u>CLAIM</u> $\quad \mathrm{im}(\phi) = \phi(G) = \{\phi(x) \mid x \in G\} \leq G'$

warning

It is not in general true that $\phi(G) \trianglelefteq G'$

▶ $G' = S_3$

$\phi : G \longrightarrow S_3$ $\qquad G = \left\{ \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix} \right\}$

$\phi(g) = g$ $\qquad$ (Previous example)

$\phi(G) = G \leq S_3$

$G \ntrianglelefteq S_3$

<u>PROOF</u>

$1 = \phi(1) \in \mathrm{im}(\phi)$

Let $x, y \in \mathrm{im}(\phi) \Rightarrow x = \phi(\alpha)$, for some $\alpha, \beta \in G$
$y = \phi(\beta)$

$xy^{-1} = \phi(\alpha)\phi(\beta)^{-1} = \phi(\alpha\beta^{-1}) \Rightarrow xy^{-1} \in \mathrm{im}(\phi)$.

FIRST ISOMORPHISM THEOREM

Let $\phi : G \longrightarrow G'$ be a homomorphism.

Then since $\ker\phi \trianglelefteq G$, we can form

$G/\ker\phi$ and $\mathrm{im}\,\phi$

and $\tilde{\phi} : G/\ker\phi \longrightarrow \mathrm{im}\,\phi = \phi(G)$

$\tilde{\phi}(g \ker\phi) = \phi(g)$

<u>CLAIM</u>

$\tilde{\phi}$ is an isomorphism.

<u>PROOF</u>

• $\tilde{\phi}$ is well-defined. $\qquad a = b \Rightarrow \tilde{\phi}(a) = \tilde{\phi}(b)$

$x \ker\phi = y \ker\phi$
$\Leftrightarrow$
$y^{-1}x \in \ker\phi$

$\Rightarrow \phi(y^{-1}x) = 1$

$\Rightarrow \phi(y)^{-1} \cdot \phi(x) = 1 \Rightarrow \phi(x) = \phi(y)$

$\Rightarrow \tilde{\phi}(x\ker\phi) = \tilde{\phi}(y\ker\phi)$

So $\tilde{\phi}$ is well-defined.

• $\tilde{\phi}$ is a homomorphism.

$\tilde{\phi}(x\ker\phi \cdot y\ker\phi) \overset{\text{def of } G/\ker\phi}{=} \tilde{\phi}(xy\ker\phi) \overset{\text{def of } \tilde{\phi}}{=} \phi(xy) \overset{\text{homomorphism}}{=} \phi(x)\phi(y)$

$= \tilde{\phi}(x\ker\phi)\tilde{\phi}(y\ker\phi)$ $\quad$ def of $\tilde{\phi}$

$\therefore \tilde{\phi}$ is a homomorphism.

• $\tilde{\phi}$ is injective.

Enough to show that

$\ker\tilde{\phi} = \{1_{G/\ker\phi}\} = \{\ker\phi\}$ ✓

Let $x\ker\phi \in \ker\tilde{\phi}$

then $\tilde{\phi}(x\ker\phi) = 1$ $\quad$ def of $\tilde{\phi}$

$\phi(x)$

$\Rightarrow$ Def of $\ker\phi$ $\quad x \in \ker\phi \Rightarrow x\ker\phi = \ker\phi$

$\phi(x) = 1_G \qquad = 1_{G/\ker\phi}$

$\Rightarrow \ker\tilde{\phi} = \{1_{G/\ker\phi}\}$

So $\tilde{\phi}$ is injection.

$\tilde{\phi}$ is surjective. Let $\alpha \in \operatorname{im}\phi$

$\Rightarrow \alpha = \phi(x) = \tilde{\phi}(x\ker\phi)$ for some $x \in G$

$\alpha \in \operatorname{im}(\tilde{\phi})$, $\therefore \tilde{\phi}$ is surjective.

Observation

Let $H \trianglelefteq G$

$\pi : G \twoheadrightarrow G/H$ canonical projection

$\pi(g) = gH$

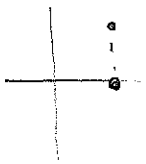
$\hookrightarrow$ inj

$\xrightarrow{\sim}$ bijective

<u>CLAIM</u> π is a homom + surj = epimorphism

<u>PROOF</u>

- $\pi(g_1 g_2) = g_1 g_2 H = g_1 H g_2 H = \pi(g_1)\pi(g_2)$
  $\Rightarrow \pi$ is a homom.
- Let $gH \in G/H$ be an arbitrary element
  $\pi(g) = gH$
  $\operatorname{im}(\pi) = G/H$



$\mathbb{R}^2$ -
H = y-axis
G/H = x axis

Observation

Let $\phi : G \to G'$ be a homomorphism.

$$\begin{tikzcd}
\ker\phi \trianglelefteq G \arrow[r, "\phi"] \arrow[d, "\pi"] & G' \\
G/\ker\phi \arrow[r, "\sim", "\tilde{\phi}"'] & \operatorname{im}\phi \arrow[u, hook, "i"']
\end{tikzcd}$$

i for inclusion (maps to itself)

COMMUTATIVE

Recall

- Suppose $N \trianglelefteq G$
  Then there is a natural homom.
  $\pi : G \twoheadrightarrow G/N$ surjection

Observation

Let $N \trianglelefteq G$

then there is a 1-1 correspondence

$X = \left\{\begin{matrix}\text{subgroups} \\ \text{of } G/N\end{matrix}\right\} \longleftrightarrow \left\{\begin{matrix}\text{subgroups of} \\ G \text{ which contain} \\ N\end{matrix}\right\} = Y$

since $N \trianglelefteq G$, $gN = Ng$, $\forall g \in G$
$\Rightarrow hN = Nh$, $\forall h \in H$
$\Rightarrow N \trianglelefteq H$

$H/N \leq G/N \qquad \boxed{\alpha(H) = H/N} \qquad N \trianglelefteq H \leq G$

$\bar{H} \leq G/N \qquad \pi^{-1}(\bar{H})$

β

$\alpha : Y \to X$ is a function.

$\alpha(H) = H/N \subseteq G/N$

WANT • $1_{G/N} = 1.N \in H/N$

• suppose $xN \in H/N$ and $yN \in H/N$

$\Rightarrow x, y \in H$

$\Rightarrow y^{-1}x \in H \Rightarrow y^{-1}xN \in H/N$

$H \leq h$

$\Rightarrow (yN)^{-1}xN \in H/N$

$\therefore H/N \leq G/N$

$\ulcorner a \in H$
$aN \in H/N \lrcorner$

$\beta : X \to Y$ is a function.

Let $\bar{H} \in X \Leftrightarrow \bar{H} \leq G/N$

• $\beta(\bar{H}) := \pi^{-1}(\bar{H})$ • $\pi : G \to G/N$

$\pi : G \twoheadrightarrow G/N$

Claim $N \leq \pi^{-1}(\bar{H}) \leq G$

Proof • $1 \in \pi^{-1}(\bar{H})$, since $\pi(1) = 1 \in \bar{H}$

since $\bar{H} \leq G/N$

since π is a homom

• Suppose $x, y \in \pi^{-1}(\bar{H})$

$\Rightarrow \pi(x), \pi(y) \in \bar{H} \Rightarrow \pi(y)^{-1}\pi(x) \in H$

$H \leq G/N$ $\quad \| \ \pi$ is homom

$\pi(y^{-1}x)$

$\Rightarrow y^{-1}x \in \pi^{-1}(\bar{H}) \quad \therefore \pi^{-1}(\bar{H}) \leq G$

$N = \ker \pi = \pi^{-1}(\{1\}) \equiv \pi^{-1}(H)$

In order to finish the proof, we need to show

(i) $\beta \circ \alpha = id$

(ii) $\alpha \circ \beta = id$

"$\beta \circ \alpha = id$ ise $H = S$ olmalı."

(i) $(\beta \circ \alpha)(H) = \beta(\alpha(H))$

$= \beta(H/N)$

• $= \pi^{-1}(H/N) = \{x \in G \mid \pi(x) \in H/N\}$

$= \{x \in G \mid xN \in H/N\}$

$= \{x \in G \mid \exists h \in H \text{ s.t. } xN = hN\}$

$= \{x \in G \mid \exists h \in H \ \ h^{-1}x \in N\} = S$

I claim that

$S = H$

a) $S \subseteq H$.

Let $x \in S \Rightarrow \exists h \in H, h^{-1}x \in N \subseteq H$

$\Rightarrow \boxed{x \in H}$

since $H \leq G$

b) $H \subseteq S$

Let $h \in H$, $h^{-1}h = 1 \in N$

$\Rightarrow h \in S$

So $\beta \circ \alpha = id$

(ii) $\alpha \circ \beta = id$

$(\alpha \circ \beta)(\bar{H})$

$= \alpha(\beta(\bar{H}))$

$= \alpha(\pi^{-1}(\bar{H}))$

$= \pi^{-1}(\bar{H})/N$

$\pi : G \longrightarrow G/H$

$\quad\quad\quad\quad\quad\quad\quad\quad \rotatebox{90}{$\in$}$

$\quad\quad\quad\quad\quad\quad\quad\quad \bar{H}$

CLAIM

$\pi^{-1}(\bar{H})/N = \bar{H}$

PROOF

First $N \trianglelefteq \pi^{-1}(\bar{H})/N \leq G$

(a) $\pi^{-1}(\bar{H})/N \subseteq \bar{H}$

Let $xN \in \pi^{-1}(\bar{H})/N$

$\left[\begin{array}{l} \Rightarrow \exists y \in \pi^{-1}(\bar{H}) \text{ s.t } xN = yN \\ \Rightarrow \exists y \in \pi^{-1}(\bar{H}) \text{ s.t } y^{-1}x \in N \end{array}\right]$

Let $xN \in \pi^{-1}(\bar{H})/N \subseteq \bar{H}$ s.t. $x \in \pi^{-1}(\bar{H})$

$\Rightarrow \pi(x) \in \bar{H}$

$\Rightarrow xN \in \bar{H}$

$\pi(x) = xN$

(b) $\bar{H} \subseteq \pi^{-1}(\bar{H})/N$,

Note $\bar{H} \leq G/N$

Let $xN \in \bar{H} \Rightarrow x \in \pi^{-1}(\bar{H}) \Rightarrow xN \in \pi^{-1}(\bar{H})/N$

$\pi(x) = xN$

▶ Let us try to find all subgroups of $(\mathbb{Z}, +)$ → cyclic

also $(\mathbb{Z}_n, +)$ cyclic

$\boxed{n\mathbb{Z} : \text{cyclic}}$

First of all, if $n \in \mathbb{N}$

then $n\mathbb{Z} = \{nz \mid z \in \mathbb{Z}\} \leq \mathbb{Z}$

CLAIM

if $H \leq \mathbb{Z}$

then $H = n\mathbb{Z}$, for some unique $n \in \mathbb{N}$

warning

$2\mathbb{Z} = (-2)\mathbb{Z}$

but $-2 \notin \mathbb{N}$

PROOF $H \neq \{0\}$

Let $|H|^* = \{|h| \mid h \in H\} \setminus \{0\} \subseteq \mathbb{N}$

Let n be the smallest element of $|H|^*$

We want to show that $H = n\mathbb{Z}$.

(a) $H \subseteq n\mathbb{Z}$

(b) $n\mathbb{Z} \subseteq H$

(b) $n \in |H|^*$

$\Rightarrow \exists h \in H$ s.t.

$n = |h|$

$\Rightarrow n = h$ or $n = -h$

for some $h \in H$

$\Rightarrow n \in H$

$\overbrace{n + n + \dots}^{m \text{ times}} \in H$

and

$\underbrace{(-n) + (-n) + \dots}_{m \text{ times}} \in H$

(a) Let $h \in H \Rightarrow |h| \in |H|^*$

$|h| = n.q + r$, for some $q \in \mathbb{N}$

and

$0 \leq r < n$.

$\Rightarrow r = (\underset{\in H}{|h|} - \underset{\in H}{n.q}) \in H$

$\Rightarrow r = 0$ or $r \in |H|^* \Rightarrow r = 0$ or $n = \min(|H|^*) \leq r$

Bunun kesulu

ama buna ters düşüyor.

$\Rightarrow |h| = n.q \Rightarrow n \in n\mathbb{Z}$

Observation

Since $(\mathbb{Z}, +)$ is abelian,
all the subgroups of $n\mathbb{Z}$ are normal.

$$\varphi : \mathbb{Z} \longrightarrow \mathbb{Z}_n$$

$\varphi(a) = \bar{a}$

$\varphi(a+b) = \overline{a+b} = \bar{a} + \bar{b} = \varphi(a) + \varphi(b)$

$\ker \varphi = n\mathbb{Z}$

$\mathbb{Z}/n\mathbb{Z} \xrightarrow{\sim} \mathbb{Z}_n$ First isomorphism

DEF: G is cyclic if $\exists g \in G$

s.t. $\langle g \rangle = G$

$\{g^n \mid n \in \mathbb{Z}\}$

Remark: If G is a group and $X \subseteq G$

similar to subspace spanned by a set.

$$\langle X \rangle = \{x_1^{\varepsilon_1}, \dots, x_n^{\varepsilon_n} \mid x_1, \dots, x_n \in X, \ \varepsilon_1, \dots, \varepsilon_n \in \{\pm 1\}\}$$

the subgroup generated by X

$$\subseteq G$$

In fact $\langle X \rangle \leq G$

$1 \in \langle X \rangle$. Let $x \in X$, $1 = x^1 x^{-1} \in \langle X \rangle$

<u>If</u> $\alpha, \beta \in \langle X \rangle$ <u>then</u>

$\alpha = x_1^{\varepsilon_1} \dots x_n^{\varepsilon_n}$ and $\beta = x_{n+1}^{\varepsilon_{n+1}} \dots x_{n+m}^{\varepsilon_{n+m}}$

where $x_1, \dots, x_{m+n} \in X$ and $\varepsilon_i \in \{\pm 1\}$

$$\alpha^{-1}\beta = (x_1^{\varepsilon_1} \dots x_n^{\varepsilon_n})^{-1} x_{n+1}^{\varepsilon_{n+1}} \dots x_{n+m}^{\varepsilon_{n+m}}$$

$$= x_1^{-\varepsilon_1} \dots x_n^{-\varepsilon_n} \cdot x_{n+1}^{\varepsilon_{n+1}} \dots x_{n+m}^{\varepsilon_{n+m}}$$

$$\in \langle X \rangle$$

If $X = \{g\}$

$\langle X \rangle = \{g^{\varepsilon_1} \dots g^{\varepsilon_n} \mid \text{where } \varepsilon_i \in \{\pm 1\}\}$

$= \{g^n \mid \text{for some } n \in \mathbb{Z}\}$

$= \langle g \rangle$

▶ Let $X = \left\{ \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix} \right\}$

Find $\underbrace{\langle X \rangle}_{\left\langle \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix} \right\rangle} \leq S_3$

DEF: - Let $g \in G$, then if $|G| < \infty$

then $|\langle g \rangle| \mid |G|$ By Lagrange.

- Suppose that G is an arbitrary group and let $g \in G$.

Then there are two possibilities.

(i) $|\langle g \rangle| < \infty$

(ii) $|\langle g \rangle|$ is not finite.

(i) If $|\langle g \rangle| < \infty$

Let $n = |\langle g \rangle|$

<u>CLAIM</u> $n = \min\{m \mid g^m = 1, m \in \mathbb{N}_{>0}\}$

PROOF

Let $n_0 = \min\{m \mid g^m = 1, m \in \mathbb{N}_{>0}\}$

(Remark: $\{m \mid g^m = 1, m \in \mathbb{N}_{>0}\} \neq \emptyset$

$\{1, g, g^2 \dots g^n\} \leq \langle g \rangle$

n elements

They cannot be all distinct (eleman sayısı olarak)

$\Rightarrow \exists \; 0 \leq a < b \leq n$

s.t $g^a = g^b$

$\Rightarrow 1 = g^{b-a}$, $b-a \in \mathbb{N}_{>0}$

$b-a \in \{m \mid g^m = 1, m \in \mathbb{N}_{>0}\}$ cannot be empty set.

WANT

$n = n_0$

PROOF

Know

(i) $g^{n_0} = 1$,

(*)(ii) if $g^m = 1$, with $1 \leq m$ then $n_0 \leq m$

$g^a = g^{n_0 q + r} = (g^{n_0})^q \cdot g^r$

$= g^r$

$\langle g \rangle = \{1, g, \dots, g^{n_0 - 1}\}$ n elements

If we can show that

$1, g, \dots, g^{n_0-1}$ are distinct then

$n = |\langle g \rangle| = |\{1, g, \dots, g^{n_0-1}\}| = n_0$

If $g^a = g^b$, for some $1 \leq a < b \leq n_0 - 1$ — en fazla derece farkı

$1 \leq b - a < n_0$

CONTRADICTS (ii)(*)

So the elements are distinct.

$|\langle g \rangle| = |g|$, this is called the order of g.

If $|\langle g \rangle| < \infty$

Then $|g| = \min\{m \mid g^m = 1, m \in \mathbb{N}_{>0}\}$

Corrolary

If $|\langle g \rangle| < \infty$, then

if $g^m = 1$, for some $m \in \mathbb{N}_{>0}$

then $|g| \mid m$

PROOF

Let $n = |g|$, $m = n \cdot q + r$ where $0 \leq r < n$

$1 = g^m = g^{nq+r} = (g^n)^q \cdot g^r = g^r$

If $r \neq 0$, then $r \in \{m \mid g^m = 1, m \in \mathbb{N}_{>0}\}$ → smallest element is n. but $r < n$ contradiction

Then r cannot be non-zero.

$r = 0$

$\Rightarrow n \mid m$

(ii) If $|\langle g \rangle| = 0$ Then we say that g has infinite order and write $|g| = \infty$

BACK TO EXAMPLE $\langle x \rangle =: H \leq S_3$

Lagrange Theorem $|H| \mid |S_3| = 6$

$x = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix} \in H$

$y = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix} \in H$

$|\langle x \rangle| = |x| = 2$ because $x^1, x^2 = 1$

$2 = |\langle x \rangle| \mid |H|$ Lagrange

$z = xy \in H$

$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix}\begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}$

$z = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}$

$z^2 = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}$

$z^3 = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix} = 1$

$|z| = 3 = |\langle z \rangle| \mid |H|$

2 divides, 3 divides

Then 6 divides $|H|$

$6 \mid |H| \rightarrow |H| = 6$

$|H| \leq 6$

$\Rightarrow H = S_3$

▶ $\left|\begin{pmatrix}1&2&3\\2&1&3\end{pmatrix}\right| = \left|\begin{pmatrix}1&2&3\\3&2&1\end{pmatrix}\right| = \left|\begin{pmatrix}1&2&3\\1&3&2\end{pmatrix}\right| = 2$

$\left|\begin{pmatrix}1&2&3\\3&1&2\end{pmatrix}\right| = \left|\begin{pmatrix}1&2&3\\2&3&1\end{pmatrix}\right| = 3$

$\left|\begin{pmatrix}1&2&3\\1&2&3\end{pmatrix}\right| = 1$

S_3 is not cyclic.

CYCLIC GROUPS

Every cyclic group is isomorphic to exactly one of the groups below

$$\mathbb{Z},\ \mathbb{Z}_n,\ n \in \mathbb{N}_{>0}$$
$$\| $$
$$\mathbb{Z}/n\mathbb{Z}$$

Suppose that G is cyclic then $G = \langle g \rangle = \{ g^n \mid n \in \mathbb{Z} \}$

$\phi : \mathbb{Z} \twoheadrightarrow G$

$\phi(n) = g^n \quad , \forall n \in \mathbb{Z}$

$\phi(n+m) = g^{n+m} = g^n \cdot g^m = \phi(n)\phi(m)$

ϕ is a homom.

First isomorphism theorem

$$\mathbb{Z}/\ker\phi \simeq \operatorname{im}\phi = G$$

$\underline{\ker\phi = n\mathbb{Z}}$, for some $n \in \mathbb{N}$

(i) If $n=0$ then $\ker\phi = \{0\}$ and ϕ is $\boxed{\text{injective}}$

$$\mathbb{Z} \xrightarrow{\sim} G, \quad |\mathbb{Z}| = \infty$$

(ii) If $n>0$ then

$$\mathbb{Z}_n \simeq \mathbb{Z}/n\mathbb{Z} \xrightarrow{\sim} G : \quad |\mathbb{Z}_n| = n$$

Recall

If G is cyclic then it is abelian.

Observations

(i) CLAIM If G is cyclic and $H \leq G$ then H is cyclic, also G/H is cyclic.

Also: $(H \trianglelefteq G)$

PROOF

We only need to prove this for $\mathbb{Z}$ and $\mathbb{Z}_n$, for some $n>0$

(a) if $G = \mathbb{Z}$ then $H = d\mathbb{Z}$ for some $d \in \mathbb{N}$.
subgroup

INFINITE ORDER $\left\{ G/H = \mathbb{Z} \right.$

$\longleftarrow$ If $d=0$ then $H = \{0\}$, so is cyclic

If $d>0$ then $H \xrightarrow{\sim} \mathbb{Z}$ ▶ $\mathbb{Z} \xrightarrow{\sim} 5\mathbb{Z}$
$a \to \frac{a}{d}$

$G/H = \mathbb{Z}/d\mathbb{Z} \simeq \mathbb{Z}_d$

(b) If $G = \mathbb{Z}_n$

$\phi : \mathbb{Z}/n\mathbb{Z} \xrightarrow{\sim} \mathbb{Z}_n$

If $\bar{H} \leq \mathbb{Z}_n$, then we proved that $\bar{H} = H/n\mathbb{Z}$ for some $n\mathbb{Z} \leq H \leq \mathbb{Z}$

$H = d\mathbb{Z}$ for some $d \in \mathbb{N}$

$n = dn'$

$\bar{H} = d\mathbb{Z}/n\mathbb{Z} \leftarrow\!\!\shortmid \mathbb{Z}/n\mathbb{Z}$

<u>CLAIM</u> $\mathbb{Z}/n\mathbb{Z} \xrightarrow{\sim} d\mathbb{Z}/n\mathbb{Z}$

$a \longrightarrow da$

$$\begin{tikzcd} \mathbb{Z} \arrow[r] \arrow[dr, "\varphi"'] & d\mathbb{Z} \arrow[d] \\ & d\mathbb{Z}/n\mathbb{Z} \end{tikzcd}$$

$\alpha \in \ker \varphi \Leftrightarrow d\alpha + n\mathbb{Z} = 0 + n\mathbb{Z}$

$\Leftrightarrow d\alpha \in n\mathbb{Z}$

$\Leftrightarrow n \mid d\alpha \Leftrightarrow \left(\frac{n}{d} \mid \alpha\right)$

$\Leftrightarrow n' \mid \alpha \Leftrightarrow \alpha \in n'\mathbb{Z}$

$n = dn'$

$\ker \varphi = n'\mathbb{Z}$ (1st isomorphism)

$\mathbb{Z}/n'\mathbb{Z} \xrightarrow{\sim} d\mathbb{Z}/n\mathbb{Z} = \bar{H}$

So $\bar{H}$ is cyclic.

$\mathbb{Z}_n/\bar{H} = \mathbb{Z}_n/d\mathbb{Z}_n = \langle 1 + d\mathbb{Z}_n \rangle$

So $\mathbb{Z}_n/\bar{H}$ is cyclic.

If G is cyclic, every subgroup and every quotient of G is cyclic as well.

(ii) Suppose that G is a cyclic group of order n. Then for every $d \mid n$ G has a unique subgroup $H \leq G$ s.t. $|H| = d$.

Moreover, if $\langle g \rangle = G$ then $H = \langle g^{n/d} \rangle$

First of all if $H \leq G$ then by Lagrange, $|H| \mid |G| = n$

Suppose $d \mid n$

Let $H = \langle g^{n/d} \rangle$

$|H| = |\langle g^{n/d} \rangle| = |g^{n/d}|$

$= \min \{ a \mid \underbrace{(g^{n/d})^a = 1}_{} \} = \min \{ a \mid g^{\frac{n \cdot a}{d}} = 1 \}$

$g^n = 1$

$= \min \{ a \mid n = |g| \mid \left(\frac{n \cdot a}{d}\right) \} = \min \{ a \mid n \mid \left(\frac{n \cdot a}{d}\right) \}$

$= \min \{ a \mid d \mid a \} = a$

Suppose $H' \leq G$ s.t. $|H'| = d$. Since $H' \leq G$, H' is cyclic.

$\therefore \exists h' \in G = \langle g \rangle$ s.t. $\langle h' \rangle = H'$

$\Rightarrow \exists h' = g^a$, for some a s.t. $\langle g^a \rangle = \langle h' \rangle = H'$

$d = |H'| = |\langle g^a \rangle| = |g^a| = \frac{n}{(n,a)}$ ▲CLAIM

Let $\alpha = \gcd(a, n) = (a, n)$

$a = \alpha \cdot a'$, $\alpha(a', n') = 1$

$n = \alpha \cdot n'$

$\therefore (n, a) = \frac{n}{d}$

▲ <u>CLAIM</u>

$|g^a| = n' = \frac{n}{(a,n)} = \frac{|g|}{(a, |g|)}$

Proof of the claim

$$|g^a| = \min\{b \mid (g^a)^b = 1\}$$
$$= \min\{b \mid g^{ab} = 1\}$$
$$= \min\{b \mid n \mid ab\}$$
$$= \min\{b \mid (\alpha n') \mid (\alpha a' b)\}$$
$$= \min\{b \mid n' \mid a' b\}$$
$$= \min\{b \mid n' \mid b\} = n' = \frac{n}{(n,a)}$$

$((n', a') = 1)$ ∎

Recall (Euclidean algorithm)

if $a, b \in \mathbb{Z}$ and $d = (a,b)$
$\exists x, y \in \mathbb{Z}$
$ax + by = d$

$\exists x, y \in \mathbb{Z}$, s.t.

⊛ $nx + ay = n/d$

WANT $\langle g^a \rangle = \langle g^{n/d} \rangle$

PROOF

$$g^{n/d} \underset{⊛}{=} g^{nx+ay} = (g^n)^x \cdot g^{ay} = (g^a)^y$$
$$g^n = 1$$

$\Rightarrow g^{n/a} \in \langle g^a \rangle \quad \in \langle g^a \rangle$

$\Rightarrow H = \langle g^{n/a} \rangle \subseteq \langle g^a \rangle = H'$

But $|H| = |H'| = d$

$\underset{H \subseteq H'}{\Longrightarrow} H = H'$

(iii) Let G be a finite group such that $|G| = p$ (prime number) then G is a cyclic group

Hence $G \xrightarrow{\sim} \mathbb{Z}_p$

PROOF of (iii) $g \in G \setminus \{1\}$, $1 \neq |\langle g \rangle| \mid |G| = \text{prime}$

$\hookrightarrow$ Lagrange

$$\underset{p \text{ is prime}}{\Longrightarrow} |\langle g \rangle| = p = |G| \Rightarrow \langle g \rangle = G$$

(iv) Suppose that G is cyclic and $|G| = n$

Q: How many generators does G have

(!) i.e $|\{g \mid \langle g \rangle = G\}| = ?$

▶ $\mathbb{Z}_6$ has two generators.

$\langle 1 \rangle \quad \langle 5 \rangle$

DEF: Let $\varphi : \mathbb{N}_{>0} \longrightarrow \mathbb{N}_{>0}$

be defined as $\varphi(n) := |\{m \mid 0 \leq m < n, (m,n) = 1\}|$

$\varphi(p) = p - 1$

$\hookrightarrow$ if p is prime.

$\boxed{1}$	$\varphi(1) = 1$	$\varphi(5) = 4$	$\boxed{1}\boxed{2}\boxed{3}\boxed{4}$
$\boxed{1}$	$\varphi(2) = 1$	$\varphi(6) = 2$	$\boxed{1}\boxed{5}$
$\boxed{1}\boxed{2}$	$\varphi(3) = 2$		
$\boxed{1}\boxed{2}$	$\varphi(4) = 2$		

$\varphi(p^n) = p^n - p^{n-1} = (p-1)\, p^{n-1}$

prime

(!) $G = \langle g \rangle$

$$\left| \{ h \mid \langle h \rangle = G \} \right| = \left| \{ g^a \mid \langle g^a \rangle = G, \ 0 \le a < n \} \right|$$

$$= \left| \{ g^a \mid |\langle g^a \rangle| = |G| = n \} \right|$$

$$= \left| \{ a \mid \tfrac{n}{(n,a)} = n, \ 0 \le a < n \} \right|$$

Claim $|g^a| = \dfrac{n}{(a,n)}$

$$= \left| \{ a \mid (n,a) = 1, \ 0 \le a < n \} \right| = \varphi(n)$$

$\square$

Recall

$n \in \mathbb{N}_{>0}$, $\varphi(n) = |\{ r \mid 0 \le r < n, (r,n) = 1 \}|$

$\varphi(n) = \#$ of generators of $\mathbb{Z}_n \cong \mathbb{Z}/n\mathbb{Z}$

$\varphi(p^k) = p^k - p^{k-1}$

DIRECT PRODUCT OF GROUPS

Let G_1, G_2 be two groups, we define a group structure on $G_1 \times G_2 = \{ (g_1, g_2) \mid g_1 \in G_1 \text{ and } g_2 \in G_2 \}$

$(g_1, g_2)(g_1', g_2') = (g_1 g_1', g_2 g_2')$

Direct product $G_1 \times G_2$ is another group

e.g. (e_{G_1}, e_{G_2}) is the identity of $G_1 \times G_2$

$(g_1, g_2)^{-1} = (g_1^{-1}, g_2^{-1})$

▶ $V = \mathbb{Z}_2 \times \mathbb{Z}_2 = \{ (0,0), (0,1), (1,0), (1,1) \}$ ABELIAN GROUP BUT NOT CYCLIC

+	(0,0)	(0,1)	(1,0)	(1,1)
(0,0)	(0,0)			
(0,1)	(0,1)			
(1,0)				
(1,1)				(0,0)

If $\alpha \in V \setminus \{0,0\}$, then $|\alpha| = 2$

$|V| = 4$

$a^2 = I, \forall a$

Later we will show that up to isomorphism there are only two groups of order 4 : $\mathbb{Z}_4 \to$ cyclic group of order 4.

$\mathbb{Z}_2 \times \mathbb{Z}_2 (= V) \to$ Klein 4 group

CLAIM If $(n,m) = 1$, $\varphi(nm) = \varphi(n)\varphi(m)$

PROOF $\psi : \mathbb{Z}_{nm} \longrightarrow \mathbb{Z}_n \times \mathbb{Z}_m$

$a \longrightarrow (a,a)$

$$\psi(a+b) = (a+b, a+b) = (a,a) + (b,b) = \psi(a) + \psi(b)$$

ψ is a homomorphism.

$a \in \ker \psi \iff (a,a) = (0,0)$ in $\mathbb{Z}_n \times \mathbb{Z}_m$

$\iff n \mid a, \ m \mid a$

$\underset{(n,m)=1}{\iff} nm \mid a$

$\iff a = 0$ in $\mathbb{Z}_{nm}$

$\Rightarrow \psi : \mathbb{Z}_{nm} \hookrightarrow \mathbb{Z}_n \times \mathbb{Z}_m$

$\Rightarrow |\mathbb{Z}_{nm}| = nm = |\mathbb{Z}_n||\mathbb{Z}_m|$

$\Rightarrow \psi : \mathbb{Z}_{nm} \xrightarrow{\sim} \mathbb{Z}_n \times \mathbb{Z}_m$ (if $(n,m) = 1$)

$\varphi(nm) = \#$ of generators of $\mathbb{Z}_{nm}$

$= \#$ of generators of $\mathbb{Z}_n \times \mathbb{Z}_m$

<u>CLAIM</u> (a,b) is the generator of $\mathbb{Z}_n \times \mathbb{Z}_m$

$\Leftrightarrow (a,n)=1$ and $(b,m)=1$

<u>PROOF</u> (a,b) is a generator of $\mathbb{Z}_n \times \mathbb{Z}_m$

$\Leftrightarrow |(a,b)| = nm$

$|(a,b)| =$ the smallest r s.t $(ra,rb)=(0,0)$

$=$ // // $n \mid ra$, $m \mid rb$

$=$ // // $\frac{n}{(n,a)} \mid r$ and $\frac{m}{(m,b)} \mid r$

$= \operatorname{lcm}\left(\frac{n}{(n,a)}, \frac{m}{(m,b)}\right)$

$= \frac{n.m}{(n,a)(m,b)}$

(a,b) is a generator of $\mathbb{Z}_n \times \mathbb{Z}_m$

$\Leftrightarrow \frac{nm}{(n,a)(m,b)} = nm$

$\Leftrightarrow (n,a)(m,b)=1$

$\Leftrightarrow (n,a)=1 \quad (m,b)=1$ $\square$

$\#$ of generators of $\mathbb{Z}_n \times \mathbb{Z}_m = \varphi(n)\,\varphi(m)$

$= \#$ of generators of $\mathbb{Z}_{nm} = \varphi(nm)$

$\square$

$\varphi(p_1^{r_1} \cdots p_k^{r_k}) = \varphi(p_1^{r_1}) \cdots \varphi(p_k^{r_k})$

$= (p_1^{r_1} - p_1^{r_1-1}) \cdots (p_k^{r_k} - p_k^{r_k-1})$

PERMUTATION GROUPS

$S_n = \{\sigma \mid \sigma : \{1,2,\dots n\} \to \{1,2,\dots n\}$ is a bijection$\}$

$(S_n, \circ)$ is a group.

$$\begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \circ \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 2 & 1 & 4 \end{pmatrix}$$

CAYLEY'S THEOREM:

If G is a finite group of order n, then G is isomorphic to a subgroup of S_n.

<u>PROOF</u> Let G be a group. Define a homomorphism

$\varphi : G \to \operatorname{Perm}(G) = S_n$

Let $g \in G$ $\varphi(g)(x) = gx, \forall x \in G$

<u>WANT</u>

$\varphi(hg) = \varphi(h) \circ \varphi(g)$

<u>CLAIM</u>

$\varphi(g) \in \operatorname{Perm}(G)$

<u>PROOF</u>

- $\varphi(g)$ <u>is injective</u>

  Because if $\varphi(g)(x) = \varphi(g)(y)$

  $\Rightarrow gx = gy$

  $\Rightarrow x = y$

- $\varphi(g)$ is surjective

PROOF for WANT

$$(\varphi(h)\circ\varphi(g))(x)$$
$$= \varphi(h)(\varphi(g)(x)) = \varphi(h)(gx)$$
$$= h(gx) = (hg)(x) = \varphi(hg)(x) \qquad \square$$

- $g \in \ker\varphi \Rightarrow \varphi(g) = id$
$$\Rightarrow \varphi(g)(1) = id(1)$$
$$\Rightarrow g = 1$$

φ is injective.

$$\Rightarrow G \hookrightarrow \mathrm{Perm}(G) \simeq S_n$$

▶ $(1\ 2\ 3) \in S_5$ is the permutation

$$(1\,2\,3) = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 3 & 1 & 4 & 5 \end{pmatrix}$$

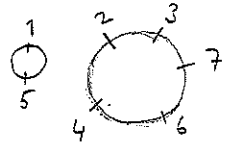
▶ $(1\,3)(1\,2) = (1\,2\,3)$

▨ 2-cycle is called a <u>transposition</u>

▶ $(3\,4)(1\,2)$ cannot be expressed as a cycle.

We will see that we can write every permutation in S_n as a product of disjoint cycles. This decomposition is unique up to ordering $\Longrightarrow$ order does not matter.

▶ $\alpha = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 5 & 3 & 7 & 2 & 1 & 4 & 6 \end{pmatrix}$



Order: $2 \cdot 5 = 10$

▶ $(1\,9\,5)(2\,7\,4\,3\,6)(8)$

disjoint since $\{1,9,5\} \cap \{2,7,4,3,6\} = \emptyset$

$\{1,9,5\}, \{2,7,4,3,6\}, \{8\}$ equivalence classes for $\sim_\sigma$

Suppose that we are given $\sigma \in S_n$

Define an equivalence relation on $\{1,2,\dots,n\}$ s.t.

$$a \sim_\sigma b \Leftrightarrow \exists\, i \in \mathbb{Z} \text{ s.t. } \sigma(a) = b$$

Reflexive
Symmetric
Transitive

The equivalence relation decomposes $\{1,2,\dots,n\}$ into disjoint equivalence classes, i.e.

$$\{1,2,\dots,n\} = \{i_1,\dots,i_k\} \dot\cup \dots \dot\cup \{j_1,\dots,j_L\}$$

disjoint union

CLAIM

$\sigma|_{\{i_1,\dots,i_k\}}$ is a cycle. Let us define

$i_2 = \sigma(i_1)$
$i_3 = \sigma(i_2)$
$\vdots$
$i_k = \sigma(i_{k-1})$
$i_1 = \sigma(i_k)$

$$\sigma = (i_1\, i_2 \dots i_k) \cdots (j_1\, j_2 \cdots j_L)$$

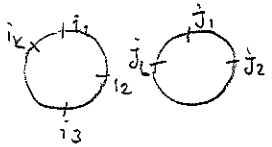
disjoint decomposition $\square$

▶ $\{1,2,3,4\}$
$\sigma(1) = 3$
$\sigma(2) = 2$
$\sigma(3) = 4$
$\sigma(4) = 1$

First element : 1
$= 2^{nd}$ " : 3
3rd " : 4
4th " : 1

suppose we have a decomposition

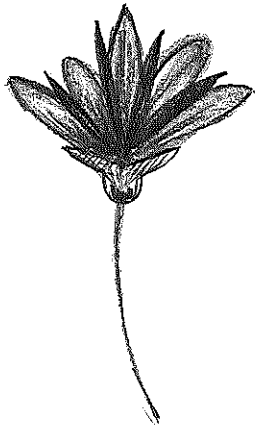
$\sigma = \gamma_1 \cdots \gamma_n$ s.t. $\gamma_1, \dots, \gamma_r$ are disjoint cycles.



If say

$v_1 = (i_1, \dots, i_k)$ then $\{i_1, \dots, i_k\}$ is one of the equivalence classes for $\sim_\sigma$

Similarly, for the other cycles.



Observation

Suppose $g, h \in G$.

s.t. $gh = hg$

$\langle g \rangle \cap \langle h \rangle = \{1\}$

Also suppose $|g|, |h| < \infty$

then $|gh| < \infty$ and $|gh| = \mathrm{Lcm}(|g|, |h|)$

Warning

The statement is not necessarily true if $gh \neq hg$

$\underset{g}{(1\,2)}\underset{h}{(1\,3)}$ $\qquad gh = (1\,3\,2)$

$|g| = |h| = 2 \qquad |gh| = 3$

$\langle g \rangle \cap \langle h \rangle = \{1\}$

PROOF Let $a = |g|$

$b = |h|$

$c = \mathrm{lcm}(a, b)$

WANT $|gh| = c$

PROOF

$$(gh)^c = \underbrace{(gh)(gh)\cdots(gh)}_{c \text{ times}}$$

$$= \underbrace{g \cdot g \cdot g \cdots g}_{c \text{ times}} \underbrace{h \cdots h}_{c \text{ times}} = g^c h^c$$

$$= (g^a)^{\frac{c}{a}} (h^b)^{\frac{c}{b}}$$

$$= 1^{\frac{c}{a}} 1^{\frac{c}{b}} = 1$$

so $|gh| \mid c$ (1)

Suppose that

$$(gh)^n = 1$$

$$\underset{gh=hg}{=} g^n h^n$$

$$\Rightarrow g^n = h^{-n} \in \langle g \rangle \cap \langle h \rangle = \{1\}$$

$$\Rightarrow g^n = h^{-n} = 1$$

$$\Rightarrow a \mid n,\ b \mid n \Rightarrow c \mid n \qquad (2)$$

$c = \mathrm{lcm}(a, b)$

$(1) + (2) \Rightarrow |gh| = c$

Observation

(i) $|(i_1 \dots i_k)| = k$

(ii) If $\sigma = \gamma_1 \dots \gamma_r$ is a product of disjoint cycles.

Then $|\sigma| = \operatorname{lcm}(|\gamma_1|, \dots, |\gamma_r|)$
$= \operatorname{lcm}(k_1, \dots, k_r)$

PROPOSITION

Every permutations $\sigma \in S_n$ can be written as a product of (not necessarily disjoint) transpositions.

PROOF

Let $\gamma \in S_n$ we can write σ as a product of cycles

$\sigma = (i_1 \dots i_k) \dots (j_1, \dots, j_k)$

Enough to show that every cycle can be written as a product of transpositions.

$(i_1 \dots i_k)$
$= (i_1 i_k) \dots (i_1 i_3)(i_1 i_2)$

This decomposition is not unique. Moreover, the number of r ($\sigma = \tau_1 \tau_2 \dots \tau_r$) is not unique either.

However we will show that $(-1)^r$ is unique.

In other words, if

$\sigma = \tau_1, \dots \tau_r$
$= \gamma_1 \dots \gamma_j$

where τ_i, γ_j are transpositions.
then $(-1)^r = (-1)^s$, i.e $2 \mid (r-s)$ even

$(1\,2) = (12)(23)(32)$

We will define a homomorphism

$$\varepsilon : S_n \longrightarrow \{\pm 1\}$$

group under multiplication

Let

$$\delta(x_1 \to x_n) = \prod_{1 \le i \le j \le n} (x_i - x_j)$$

- if $n=2$
  $\delta(x_1, x_2) = x_1 - x_2$
- $n=3$
  $\delta(x_1, x_2, x_3) = (x_1 - x_2)(x_1 - x_3)(x_2 - x_3)$

Let $\sigma \in S_n$, $\varepsilon(\sigma) = \dfrac{\delta(x_{\sigma(1)}, \dots, x_{\sigma(n)})}{\delta(x_1, \dots, x_n)} \in \{\pm 1\}$

▶ $\sigma = (1,2,3)$ Bunla başla, hepsini dolaş. (all pairs)

$$\varepsilon(\sigma) = \frac{\delta(x_2, x_3, x_1)}{\delta(x_1, x_2, x_3)} = \frac{(x_2 - x_3)(x_2 - x_1)(x_3 - x_1)}{(x_1 - x_2)(x_1 - x_2)(x_2 - x_3)} = +1$$

▶ Let $\tilde{\delta}(x_1, x_2, x_3) = (x_2 - x_1)(x_1 - x_3)(x_3 - x_2)$

$$\varepsilon(\sigma) := \frac{\tilde{\delta}(x_{\sigma(1)}, x_{\sigma(2)}, x_{\sigma(3)})}{\tilde{\delta}(x_1, x_2, x_3)}$$

$$= \frac{(x_1 - x_2)(x_3 - x_1)(x_3 - x_2)}{(x_2 - x_1)(x_1 - x_3)(x_3 - x_2)} = +1$$

Lemma

Let $\gamma \in S_n$, then $\forall \sigma \in S_n$

$$\frac{\delta(x_{\sigma(1)}, \dots, x_{\sigma(n)})}{\delta(x_1, x_2, \dots, x_n)} = \frac{\delta(x_{\sigma(\gamma(1)}, \dots, x_{\sigma(\gamma(n)})}{\delta(x_{\gamma(1)}, \dots, x_{\gamma(n)})}$$

PROOF of Lemma

$$f(x_{\sigma(1)}, \dots, x_{\sigma(n)})$$

$$= \prod_{1 \le i < j \le n} (x_{\gamma(i)} - x_{\gamma(j)})$$

$$= \pm \prod_{1 \le i < j \le n} (x_i - x_j)$$

then

$$\frac{\sigma(f(x_{\gamma(1)}, \dots, x_{\gamma(n)}))}{f(x_{\gamma(1)}, \dots, x_{\gamma(n)})} = \frac{\gamma(\mp f(x_1, \dots, x_n))}{\pm f(x_1, \dots, x_n)} = \frac{\pm f(x_{\sigma(1)} \dots x_{\sigma(n)})}{\pm f(x_1, \dots, x_n)}$$

$$= + \frac{f(x_{\sigma(1)}, \dots, x_{\sigma(n)})}{f(x_1, x_2, \dots, x_n)}$$

$\square$

▶ $f(x_1, x_2, x_3) = (x_1 - x_2)(x_1 - x_3)(x_2 - x_3)$

$$\gamma = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix} \quad (x_2 - x_1)(x_2 - x_3)(x_1 - x_3)$$

Let $\gamma, \sigma \in S_n$

$$\varepsilon(\sigma\gamma) = \frac{f(x_{\sigma\gamma(1)}, \dots, x_{\sigma\gamma(n)})}{f(x_1, \dots, x_n)}$$

$$= \frac{f(x_{\sigma\gamma(1)}, \dots, x_{\sigma\gamma(n)})}{f(x_{\gamma(1)} \dots x_{\gamma(n)})} \cdot \frac{f(x_{\gamma(1)} \dots, x_{\gamma(n)})}{f(x_1, \dots, x_n)}$$

$$\underset{\text{Lemma}}{=} \frac{f(x_{\sigma(1)} \dots x_{\sigma(n)})}{f(x_1, \dots, x_n)} \cdot \frac{f(x_{\gamma(1)}, \dots, x_{\gamma(n)})}{f(x_1, \dots, x_n)}$$

$$= \varepsilon(\sigma)\varepsilon(\gamma)$$

HOMOMORPHISM

(*) Corollary

If $\tau_1, \dots, \tau_r = \Theta_1 \cdots \Theta_s$ where τ_1, Θ_s are transpositions then

$$(-1)^r = (-1)^s$$

PROOF

WANT:

If $\tau \in S_n$, and τ is a transposition then

$$\varepsilon(\tau) = -1$$

PROOF

Suppose that

$$\tau = (i\ j) \qquad f = \begin{pmatrix} \dots & i & \dots & j & \dots \\ \dots & 1 & \dots & 2 & \dots \end{pmatrix}$$

$$f\tau f^{-1} = \begin{pmatrix} 1 & 2 & 3 & 4 & \dots \\ 2 & 1 & 3 & 4 & \dots \end{pmatrix} = (12)$$

ADVANTAGE

$$\varepsilon((12))$$

$$\varepsilon(f\tau f^{-1}) = \varepsilon(f)\varepsilon(\tau)\varepsilon(f^{-1})$$

$$= \varepsilon(f)\varepsilon(\tau)\varepsilon(f)^{-1}$$

$$= \varepsilon(\tau)$$

$$\varepsilon((12)) = \frac{(x_2 - x_1)(x_2 - x_3) \cdots (x_{n-1} - x_n)}{(x_1 - x_2)(x_1 - x_3) \cdots (x_{n-1} - x_n)}$$

NOTE: ε is sign.

(*) PROOF

$$(-1)^r = \varepsilon(\tau_1) \cdots \varepsilon(\tau_r) = \varepsilon(\tau_1 \cdots \tau_r) = \varepsilon(\Theta_1 \cdots \Theta_s)$$

$$= \varepsilon(\Theta_1) \cdots \varepsilon(\Theta_s)$$

$$= (-1)^s$$

DEF

We say that σ is an even permutation (respectively odd " $\varepsilon(\gamma)=-1$)

If $\varepsilon(\gamma)=+1$

Let $\ker(\varepsilon) = A_n \trianglelefteq S_n$

called the alternating group

$$S_n/A_n \xrightarrow{\sim} \{\pm 1\}$$

First isomorphism

$$(S_n : A_n) = 2$$

▶ $A_3 = \{1, (1\,2\,3), (1\,3\,2)\}$

daima cift

$A_4 = \{1, (1\,2)(3\,4), (1\,3)(2\,4)$
$(1\,4)(2\,3), (1\,2\,3), (1\,3\,2)$
$(1\,2\,4), (1\,4\,2), (1\,3\,4), (1\,4\,3)$
$(2\,3\,4), (2\,4\,3)\}$

conjugate in S_4

$\underbrace{\sigma\, 1\, \sigma^{-1}}_{1}$

1 is conjugate to itself only.

DEF

Let G be a group then we say that g_1 and g_2 are conjugate in G, if $\exists\, \alpha \in G$ s.t.

$$\alpha^{-1} g_1 \alpha = g_2$$

equivalence relation

another notation: $g^{\alpha} = \alpha^{-1} g \alpha$

$(g^{\alpha})^{\beta} = g^{\alpha\beta}$

$(gh)^{\alpha} = g^{\alpha} h^{\alpha}$

Observation (Let $\sigma \in S_n$ — Permutation)

$$\underbrace{\sigma(i_1, \dots, i_r)\sigma^{-1}}_{\text{another notation } (i_1,\dots,i_r)^{\sigma^{-1}}} = (\sigma(i_1)\,\sigma(i_2)\,\sigma(i_3)\cdots\sigma(i_r))$$

Rest is fixed!

▶ $n=4$ $\quad \sigma = (1\,3)(2\,4) = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix}$

$$\sigma\,(1\,2\,3)\,\sigma^{-1} = (3\,4\,1)$$

We can generalize as follows

$$\sigma \overbrace{(i_{11} \cdots i_{1r_1})(i_{21} \cdots i_{2r_2}) \cdots (i_{n1} \cdots i_{nr_n})}^{A} \sigma^{-1}$$

$$\underset{\text{NOTATION}}{=} \left[(i_{11} \cdots i_{1r_1}) \cdots (i_{n1} \cdots i_{nr_n})\right]^{\sigma^{-1}}$$

$$= (i_{11} \cdots i_{1r_1})^{\sigma^{-1}} \cdots (i_{n1} \cdots i_{nr_n})^{\sigma^{-1}}$$

$$= (\sigma(i_{11}) \cdots \sigma(i_{1r_1})) \cdots (\sigma(i_{n1}) \cdots \sigma(i_{nr_n})) \quad (*)$$

If A is a disjoint cycle decomposition then so is

$(*)$

In other words, if γ is a permutation with a disjoint cycle decomposition of type $r_1, \dots, r_k$ then so is $\gamma^{\sigma^{-1}}$ for any $\sigma \in S_n$

▶ 2, 2, 3, 4

$\gamma = (3\,2)(1\,4\,5)(6\,9)(7\,8\,10\,11)$ disjoint cycle decomposition

If $\sigma \in S_n$, $\gamma^{\sigma^{-1}}$ will also have a disjoint cycle decom of type 2 2 3 4

Conversely suppose that γ_1 and γ_2 have the same disjoint cycle decomposition type.

Then is it true that γ_1 and γ_2 are conjugate, i.e. is it true that $\exists \sigma^{-1} \in S_n$, s.t.

$$\gamma_1^{\sigma^{-1}} = \gamma_2$$

$$\gamma_1 = (i_{11}, \dots, i_{1r}) \cdots (i_{k1} \cdots i_{kr})$$

$$\gamma_2 = (j_{11} \cdots j_{1r}) \cdots (j_{k1} \cdots j_{kr})$$

s.t. $i_{\alpha\beta} \neq i_{\alpha'\beta'}$ if $\alpha \neq \alpha'$ OR $\beta \neq \beta'$

also $j_{\alpha\beta} \neq j_{\alpha'\beta'}$ if $\alpha \neq \alpha'$ OR $\beta \neq \beta'$

Let be s.t

$$\sigma(i_{\alpha\beta}) = j_{\alpha\beta}$$

and σ sends the other elements arbitrarily s.t. σ is a permutation of $\{1, \dots, n\}$

$$\gamma_1^{\sigma^{-1}} = (\sigma(i_{11}) \cdots \sigma(i_{1r_1})) \cdots (\sigma(i_{k1}) \cdots \sigma(i_{kr_k}))$$
$$= (j_{11} \cdots j_{1r_1}) \cdots (j_{k1} \cdots j_{kr_k})$$
$$= \gamma_2$$

$\gamma_1, \gamma_2 \in S_n$ are conjugate in S_n
$\Leftrightarrow$ γ_1 and γ_2 have the same disjoint cycle decomposition type.

▶ Determine all the conjugacy classes (i.e. the equivalence classes of elements which are conjugate to each other) in S_4.

$\{1\}$		1
$\{(12), (13) \dots\}$		6
$\{(12)(34), \dots\}$		3
$\{(123), \dots\}$		8
$\{(1234) \dots\}$		6

▶ A_4 (Page 24)

$|A_4| = 12$

A_4 is non-abelian.

$(124)(123) \neq (123)(124)$

Let

$$V = \{1, \underbrace{(12)(34)}_{a}, \underbrace{(13)(24)}_{b}, \underbrace{(14)(23)}_{c}\}$$

CLAIM $V \trianglelefteq A_4$

$a^2 = b^2 = c^2$

$ba = c = ab$
$ac = b = ca$
$bc = a = cb$

V forms an abelian group of order 4.

$$V \simeq \mathbb{Z}_2 \times \mathbb{Z}_2$$

$\mathbb{Z}_p \times \mathbb{Z}_p$ or $\mathbb{Z}_{p^2}$ } ABELIAN

$(12)(34) \longrightarrow (0,0)$
$(13)(24) \longrightarrow (1,0)$
$(14)(23) \longrightarrow (0,1)$
$(1,1)$

Recall that by Lagrange Theorem,

if $H \leq A_4$ then

$|H| \mid |A_4| = 12$

We will show that the naive converse to Lagrange Theorem need not be true

In fact A_4 does not have a subgroup of order 6.

Suppose that

$H \leq A_4$ and $|H| = 6$

$\Rightarrow (A_4 : H) = 2 \Rightarrow H \trianglelefteq A_4$

$1 \in H$ and H has to contain an element of the form

$(i\ j\ k)$

$\longrightarrow$ Let represent it as $(1\ 2\ 3)$

$(123) \in H \Rightarrow (132) \in H$

Let $\sigma = (12)(34) \in A_4$ $\longrightarrow$ since H is normal subgroup and σ is element of A_4. ($\triangleright$ gHg^{-1})

$\sigma (123) \sigma^{-1} = (214) \in H$

$\Downarrow$

$(241) \in H$ (cancel)

since $H \trianglelefteq A_4$.

$\tau = (13)(24) \in A_4$

$\tau (123) \tau^{-1} = (341) \in H \Rightarrow (314) \in H$

$1, (123), (132), (214), (241), (341), (314) \in H$

so such a subgroup $H \leq A_4$ w/ $|H| = 6$ does not exist.

DIRECT PRODUCT OF GROUPS AND THE FUNDAMENTAL THEOREM OF FINITELY GENERATED ABELIAN GROUP.

DEF: Let $G_1, \dots, G_n$ be groups. The direct product (direct sum) of $G_1, \dots, G_n$, is the cartesian product.

$G_1 \times \dots \times G_n = \{(g_1 \dots g_n) \mid g_i \in G_i\}$

together with the operation.

$(g_1 \dots g_n) \cdot (h_1 \dots h_n)$

$= (g_1 h_1, g_2 h_2, \dots, g_n h_n)$

$\in G_1 \times \dots \times G_n$

This makes $G_1 \times \dots \times G_n$ a group.

<u>Observation</u>

If $|g_i| < \infty$ for all $1 \leq i \leq n$

$|(g_1 \dots g_n)| = \operatorname{lcm}(|g_1| \dots |g_n|)$

Let $|g_i| = r_i$ and let $r = \operatorname{lcm}(r_1, \dots, r_n)$

<u>WANT</u>

$r = |g|$

<u>PROOF</u>

$(g_1, \dots g_n)^a = 1$

$\Leftrightarrow (g_1^a \dots, g_n^a) = (1, 1, 1, \dots)$

$\Leftrightarrow g_i^a = 1$, for all $1 \leq i \leq n$

$\Leftrightarrow |g_i| \mid a$ for all $1 \leq i \leq n$

$\Leftrightarrow r_i \mid a$ for all $1 \leq i \leq n$

$\Leftrightarrow r \mid a$

$r = |(g_1 \dots, g_n)|$

▶ (i) $(2,6) \in \mathbb{Z}_4 \times \mathbb{Z}_{12}$
(ii) $(2,3) \in \mathbb{Z}_6 \times \mathbb{Z}_{15}$
(iii) $(8,10) \in \mathbb{Z}_{12} \times \mathbb{Z}_{18}$
(iv) $(3,10,9) \in \mathbb{Z}_4 \times \mathbb{Z}_{12} \times \mathbb{Z}_{15}$
(v) $(3,6,12,16) \in \mathbb{Z}_4 \times \mathbb{Z}_{12} \times \mathbb{Z}_{20} \times \mathbb{Z}_{24}$

NOT CYCLIC

(i) $|(2,6)| = \operatorname{lcm}(|2|, |6|)$
$= \operatorname{lcm}\left(\frac{4}{(4,2)}, \frac{12}{(6,12)}\right)$
$= \operatorname{lcm}(2,2) = 2$

(ii) 15

(iii) 9

(iv) 60

(v) 60

$\mathbb{Z}_{m_1} \times \mathbb{Z}_{m_2} \times \dots \times \mathbb{Z}_{m_r}$ is a cyclic group
$\Leftrightarrow (m_i, m_j) = 1$ for every $i \neq j$

PROOF

($\Leftarrow$) $|(1,1,1,1)| = \operatorname{lcm}((1)(1)\cdots(1))$
$= \operatorname{lcm}(m_1, m_2, \dots m_r)$
$= m_1 m_2 \dots m_r$
Since $(m_i, m_j) = 1$ for every $i \neq j$

$|\mathbb{Z}_{m_1} \times \mathbb{Z}_{m_2} \times \dots \times \mathbb{Z}_{m_r}| = |\mathbb{Z}_{m_1}| \cdots |\mathbb{Z}_{m_r}|$
$= m_1 \dots m_r$

$\langle (1, \dots, 1) \rangle = \mathbb{Z}_{m_1} \times \mathbb{Z}_{m_2} \dots \times \mathbb{Z}_{m_r}$
so

$\mathbb{Z}_{m_1} \times \dots \mathbb{Z}_{m_r}$ is cyclic hence
$\mathbb{Z}_{m_1} \times \mathbb{Z}_{m_2} \simeq \mathbb{Z}_{m_1 m_2 \dots m_r}$

($\Rightarrow$) If $\mathbb{Z}_{m_1} \times \dots \times \mathbb{Z}_{m_r}$ is cyclic then $\exists (a_1 \dots a_r) \in \mathbb{Z}_{m_1} \times \dots \mathbb{Z}_{m_r}$
s.t
$\operatorname{lcm}\left(\frac{m_1}{(m_1, a_1)} \dots \frac{m_r}{(m_r, a_r)}\right) = |(a_1 \dots a_r)| = |\mathbb{Z}_{m_1} \times \dots \times \mathbb{Z}_{m_r}|$

$\Rightarrow \operatorname{lcm}(m_1, \dots m_r) = m_1 \dots m_r$
$\Rightarrow (m_i, m_j) = 1$ for $i \neq j$

▶ $\mathbb{Z}_3 \times \mathbb{Z}_{77} \times \mathbb{Z}_{26} \simeq \mathbb{Z}_{3 \times 77 \times 26}$

THEOREM *Fundamental Theorem*

Let A be a finitely generated Abelian group then

A is isomorphic to a group of type

$$\underbrace{\mathbb{Z} \times \mathbb{Z} \times \dots \times \mathbb{Z}}_{r \text{ times}} \times (\mathbb{Z}_{p_1^{r_1}} \times \mathbb{Z}_{p_1^{r_2}} \times \dots \times \mathbb{Z}_{p_1^{r_{k_1}}})$$
$$\times (\mathbb{Z}_{p_2^{s_1}} \times \dots \times \mathbb{Z}_{p_1^{s_{k_2}}})$$
$$\times$$
$$\times (\mathbb{Z}_{p_\ell^{t_1}} \times \dots \mathbb{Z}_{p_\ell^{t_{k_\ell}}})$$

s.t.
p_i are prime
$p_1 < p_2 \dots < p_\ell$ and

$r_1 \leq r_2 \leq \dots \leq r_{k_1}$
$s_1 \leq s_2 \leq \dots \leq s_{k_2}$
$t_1 \leq t_2 \leq \dots \leq t_{k_\ell}$

and this decomposition is unique.

DEF r is called the rank of A (book calls it Betti number)

We can also write A in the statement as
$\mathbb{Z} \times \dots \times \mathbb{Z} \times \mathbb{Z}_{m_1} \times \mathbb{Z}_{m_2} \times \dots \times \mathbb{Z}_{m_g}$

with $m_1 | m_2 | m_3 \dots | m_g$

This is also unique

If A is abelian, and

$|A| = n < \infty$

if $m \mid n$ then

A has a subgroup of order m.

PROOF

$$A = (\mathbb{Z}_{p_1^{r_1}} \times \dots \times \mathbb{Z}_{p_1^{r_k}}) \times \times (\mathbb{Z}_{p_\ell^{t_1}} \times \dots \times \mathbb{Z}_{p_\ell^{t_{k_\ell}}})$$

$(r = 0)$

$n = |A| = p_1^{r_1} \cdots p_\ell^{r_1} \cdots p_\ell^{t_1} \cdots p_\ell^{t_{k_\ell}}$

$m \mid n \quad m = p_1^{r_1'} \cdots p_\ell^{t_1'}$

$r_1' \le r_1$

$r_2' \le r_2$

$\vdots$

$t_{k_\ell}' \le t_{k_\ell}$

$\boxed{\langle p_1^{r_1 - r_1'} \rangle \le \mathbb{Z}_{p_1^{r_1}}}$, $|\langle p_1^{r_1 - r_1'} \rangle| = p_1^{r_1'}$

$$B = \langle p_1^{r_1 - r_1'} \rangle \times \langle p_1^{r_2 - r_2'} \rangle \times \dots \times \quad \le A$$

$$|B| = p_1^{r_1'} \cdots p_\ell^{t_{k_\ell}} = m$$

$\square$

▶ Find all abelian groups up to isomorphism of order 72.

$72 = 2^3 \cdot 3^2$

$\mathbb{Z}_{2^3}$ $\quad$ $\mathbb{Z}_{3^2}$

$\mathbb{Z}_2 \times \mathbb{Z}_{2^2}$ $\quad$ $\mathbb{Z}_3 \times \mathbb{Z}_3$

$\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$

▶ Determine which abelian group is

$$A = \mathbb{Z}_6 \times \mathbb{Z}_{12} / \langle (3,4) \rangle$$

isomorphic to?

$|\langle (3,4) \rangle| = |(3,4)|$

$= \operatorname{lcm}\left(\frac{6}{(6,3)}, \frac{12}{(4,12)}\right)$

$= \operatorname{lcm}(2,3) = 6$

72 distinct pair

$$\left| \mathbb{Z}_6 \times \mathbb{Z}_{12} / \langle (3,4) \rangle \right| = \frac{6 \cdot 12}{6} = 12$$

So A is isomorphic to either $\mathbb{Z}_{2^2} \times \mathbb{Z}_3$ or $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_3$

this has an element of order 4

this does not have

Does A have an element of order 4?

if $(a,b) \in A$ in of order 4 then

$4(a,b) = 0$ in A but $2(a,b) \ne 0$ in A

$(4a, 4b) \in \langle (3,4) \rangle = \{(0,0)(3,4)(0,3)(3,0)(0,4)(3,8)\} \Rightarrow (4a,4b) = (0,0)$ $(0,8)$ or $(0,4)$

Let $(a,b) = (0,1)$ then $4(a,b) = 4(0,1) = (0,4) \in \langle (3,4) \rangle$

but $2(a,b) = 2(0,1) = (0,2) \notin \langle (3,4) \rangle$

DEF:

We say that a group G is simple if

$N \trianglelefteq G \Rightarrow N = \{1\}$ or $N = G$

▶ If $|G| = p$, then G is simple

Let $N \trianglelefteq G$

$|N| \mid |G| = p$

$\Rightarrow |N| = 1 \Rightarrow N = \{1\}$

$|N| = p \quad N = \{G\}$

DEF:

Let G be a group

$Z(G) = \{g \in G \mid gh = hg, \forall h \in G\}$

$Z(G) \leq G$

if $g \in Z(G)$ and $\alpha \in G$

$g^\alpha = \alpha^{-1} g \alpha = g \in Z(G)$

$Z(G)^\alpha = Z(G), \forall \alpha \in G$

$\Rightarrow Z(G) \trianglelefteq G$

↳ center of G.

▨ G is abelian $\Leftrightarrow Z(G) = G$

$Z(G)$ measures how close to being abelian G.

DEF: Let $g, h \in G$

$[g,h] = ghg^{-1}h^{-1} \in G$

↳ called the commutator of g and h

Note

$gh = hg \Leftrightarrow [g,h] = 1$

$[G,G] = \langle [g,h] \mid g,h \in G \rangle$ is called the commutator subgroup of G.

Claim

$[G,G] \trianglelefteq G$

PROOF

$[g,h]^\alpha = (ghg^{-1}h^{-1})^\alpha$

$= g^\alpha h^\alpha (g^\alpha)^{-1} (h^\alpha)^{-1} = [g^\alpha, h^\alpha]$

Let $x \in [G,G]$

$\Rightarrow x = [g_1,h_1]^{\varepsilon_1} \cdots [g_n,h_n]^{\varepsilon_n}$ w/ $\varepsilon_i \in \mathbb{Z}$

$x^\alpha = ([g_1,h_1]^{\varepsilon_1} \cdots [g_n,h_n]^{\varepsilon_n})^\alpha = [g_1^\alpha, h_1^\alpha]^{\varepsilon_1} \cdots [g_n^\alpha, h_n^\alpha]^{\varepsilon_n}$

$= [G,G]$

• $[G,G] \trianglelefteq G$

G is abelian $\Rightarrow [G,G] = \{1\}$

PROPOSITION

$G/[G,G]$ is abelian.

PROOF

Let $x[G,G], y[G,G] \in G/[G,G]$

WANT

$[x[G,G], y[G,G]] = 1$ in $G/[G,G]$

$\Leftrightarrow x[G,G]\, y[G,G]\, (x[G,G])^{-1} (y[G,G])^{-1} = 1[G,G]$ in $G/[G,G]$

$\Leftrightarrow [x,y] = [G,G] = 1[G,G]$ in $G/[G,G]$

$\Leftrightarrow [x,y] \in [G,G]$ □

THEOREM

A_n is simple for $n \geq 5$

Recall

Note that A_4 is not simple.

$V = \{1, (12)(34), (13)(24), (14)(23)\} \trianglelefteq A_4$

PROOF

- A_n is generated by 3-cycles.

Let $\sigma \in A_n$, we know that σ can be written as a product of an even number of transpositions.

Therefore it is enough to show that a product of two transpositions can be written as a product of 3-cycles.

CASE 1: $(ik)(ij) = 1 = (ijk)(ikj)$

CASE 2: $(ij)(ik) = (ikj)$

CASE 3: $(ij)(kl) = (ikj)(kli)$

$\therefore$ A_n is generated by 3-cycles.

- Any two 3-cycles in A_n one conjugate to each other in A_n.

Let (ijk) and $(i'j'k')$ be two 3 cycles in A_n.

Know $\exists\, \sigma \in S_n$ s.t. $(ijk)^{\sigma} = (i'j'k')$

If $\sigma \subset A_n$ then we are done.

So suppose $\sigma \notin A_n \Rightarrow \sigma$ is odd.

Let $\{r,s\} \cap \{i,j,k\} = \emptyset$

$\sigma(rs) \in A_n$

$(ijk)^{\sigma(rs)} = \left((ijk)^{\sigma}\right)^{(rs)} = (i'j'k')^{(rs)} = (i'j'k')$

$\therefore$ $(i'j'k')$ and (ijk) are conjugate in A_n $\square$

CLAIM

If $N \trianglelefteq A$ and N contains a 3-cycle $\Rightarrow N = A_n$.

PROOF

Since $N \trianglelefteq A_n$, $\forall \sigma \in A_n$

$N^{\sigma} = N \Rightarrow$

if $(ijk) \in N \Rightarrow (ijk)^{\sigma} \in N$

$\forall \sigma \in A_n$

$\Rightarrow$ since $\forall\, (i'j'k')$

$\exists\, \sigma \in A_n$ s.t.

$(i'j'k') = (ijk)^{\sigma}$

So suppose

$1 \neq N \trianglelefteq A_n$

Let $1 \neq \sigma \in N$ s.t. σ fixes the maximal # of elements in $\{1, 2, \dots, n\}$

We want to show that σ is a 3-cycle.

Look at the disjoint cycle decomposition of σ.

(i) If this dec. contains a cycle w/ at least 5 elements.

$\sigma = (1234\dots)$

$\gamma = \underbrace{(132)\left[(12345\dots)\dots\right](123)\sigma^{-1}}_{\in N}$

$\|$

$[(31245\dots)\dots]\sigma^{-1}$

$\|$

$[(31245\dots)\dots](54321\dots)(\dots$

$\dots(\dots)\cdot(22)\dots \in N$

$\neq$

I

This new element $\gamma \neq 1 \in N$ and fixes more elements than σ.

$\times$

(ii) There is an orbit in σ that contains exactly 3 elements and another orbit.

$\sigma = (123)(45\dots)\dots$

$\gamma = \left[(123)(45\dots)\right]^{(145)}\sigma^{-1} \in N$

$= (523)(\dots)\sigma^{-1} \in N$

$= \dots(33)$

$\neq 1$

γ fixes more elements than σ

(iii)

There are at least two transpositions appearing in σ:

$$\sigma = (12)(34)\cdots$$

$$1 \neq \sigma^{(125)} \cdot \sigma^{-1} \in N$$

$$\overset{\shortparallel}{(51)(34)\cdots\sigma^{-1}} \in N$$

fixes at least two more elements than σ.

$\times$

EXAM

1)(a) Let A be an abelian group w/ $H, K \leq A$ s.t. $|H|=r$, $|K|=s$, $(r,s)=1$ and H and K are cyclic. Show that A has a cyclic subgroup of order rs.

HAVE

$\langle h\rangle = H$, $\langle k\rangle = K$, $|h| = |\langle h\rangle| = |H| = r$
$|k| = |\langle k\rangle| = |K| = s$

CLAIM

$|hk| = rs$

PROOF

Need to show that two things.

(i) $(hk)^{rs} = 1$ ✓

(ii) If $(hk)^n = 1$ then $rs \mid n$.

(i) $(hk)^{rs} = h^{rs}k^{rs} = (h^r)^s \cdot (k^s)^r$

since A is abelian

$= 1^s \cdot 1^s = 1$

(ii) If $(hk)^n = 1 \Rightarrow h^n = k^{-n} \in H \cap K$

$|H\cap K| \mid |H| = r$
$|H\cap K| \mid |K| = s$ $\Longrightarrow (H\cap K) = 1$ $(r,s)=1$

$H \cap K = \{1\}$

$\Rightarrow h^n = k^{-n} \in H\cap K = \{1\}$

$\Rightarrow h^n = k^{-n} = 1$

$\Rightarrow h^n = 1$, $k^{-n} = 1$ $\Rightarrow r \mid n \Rightarrow rs \mid n$

and $s \mid n$

$(r,s)=1$

$\square$

$|\langle hk\rangle| = |hk| = rs$

$\langle hk\rangle$ is a subgroup of order n.

$|h| = r$, $|k| = s$ and $(r,s)=1$ and $hk = kh$ then $|hk| = rs$

(b) Prove that this need not be true if r and s are <u>not</u> relatively prime

Take $A = V = \mathbb{Z}_2 \times \mathbb{Z}_2$

This is abelian

$r = s = 2$

It has cyclic subgroups of order r and s. But no cyclic subgroup of order 4.

2) Prove $(\mathbb{C}^\times, \cdot)$ and $(\mathbb{R}, +)$ are not isomorphic.

Suppose that they are isomorphic then $\exists\, \phi : \mathbb{C}^\times \to \mathbb{R}$ which is an isomorphism.

$-1 \in \mathbb{C}^\times$ has order 2.

$\Rightarrow \phi(-1)$ has order 2 in $\mathbb{R}$.

$(\mathbb{R}, +)$ does not have any element of order 2 since $x + x = 0 \Rightarrow x = 0$ in $\mathbb{R}$.

3) Show that if $\sigma \in S_n$ is a cycle of odd order then so is σ^2

$$\sigma = (a_1\ a_2 \dots a_{2k+1})$$
$$\sigma^2 = (a_1 a_3 a_5 \cdots\ a_2 a_4 \cdots)$$

4) $A = (\mathbb{Z}_4 \times \mathbb{Z}_{12}) / (\langle 2 \rangle \times \langle 2 \rangle)$ as a product of cyclic groups.

$$|\langle 2 \rangle \times \langle 2 \rangle| = |\langle 2 \rangle| \cdot |\langle 2 \rangle|$$
$$= |\{0,2\}| \cdot |\{0,2,4,6,8,10\}|$$
$$= 12$$

$$|A| = \frac{4 \cdot 12}{12} = 4$$

$A \simeq \mathbb{Z}_4$ OR $\mathbb{Z}_2 \times \mathbb{Z}_2$

Let $(a,b) + (\langle 2 \rangle \times \langle 2 \rangle) \in A$
$2(a,b) + (\langle 2 \rangle \times \langle 2 \rangle)$
$= (2a, 2b) + (\langle 2 \rangle \times \langle 2 \rangle)$
$= 0 + (\langle 2 \rangle \times \langle 2 \rangle)$
So if $a \in A$, $|a| = 1$ OR 2

$$\Rightarrow A \simeq \mathbb{Z}_2 \times \mathbb{Z}_2$$

5) Is there a non-zero homomorphism

$$\phi : \mathbb{Z}_{12} \to \mathbb{Z}_5 \ ?$$

Let $\phi : \mathbb{Z}_{12} \to \mathbb{Z}_5$ be a homomorphism

$$\mathbb{Z}_{12}/\ker\phi \simeq \operatorname{im}(\phi) \leq \mathbb{Z}_5$$
$$\Rightarrow |\operatorname{im}\phi| \,\big|\, |\mathbb{Z}_5| = 5$$
$$\Rightarrow |\operatorname{im}\phi| = 1 \text{ OR } |\operatorname{im}\phi| = 5$$

$\Rightarrow$ if $|\operatorname{im}\phi| = 1$ then ϕ is the 0 homomorphism.

Suppose $|\operatorname{im}\phi| = 5$ $\quad \frac{12}{|\operatorname{im}\phi|} = |\mathbb{Z}_{12}/\ker\phi| = 5$

GROUP ACTIONS ON SETS

DEF: Let G be a group and X be a set. Then an action of G or X is a function:

$$G \times X \to X$$
$$(g, x) \longrightarrow g * x$$

s.t

(i) $e * x = x$, for every $x \in X$

(ii) for every $g_1, g_2 \in G$ and $x \in X$

$$(g_1 \cdot g_2) * x = g_1 * (g_2 * x)$$

▶ If G is a group then G acts on itself.

(i)
$$G \times G \longrightarrow G$$
$$(g, x) \longrightarrow gx$$

▶ Let $H \leq G$, then G acts on $\underbrace{\mathcal{L}_H(G)}_{\text{only a set}}$

(ii)

$$G \times \mathcal{L}_H(G) \longrightarrow \mathcal{L}_H(G)$$
$$\ell * (kH) \longrightarrow (\ell k) H$$

$e * (kH) = (ek)H = kH$ (i)✓

$(\ell_1 \ell_2) * (k \cdot H) = (\ell_1 \ell_2) kH = \ell_1 (\ell_2 k) H$
$= \ell_1 * ((\ell_2 \cdot k) \cdot H) = \ell_1 * (\ell_2 * kH)$ (ii)✓

Observation — ↳ R to plane

(i) Suppose that G acts on X. We have

$$G \times X \to X$$
$$(g, x) \to g * X$$

CLAIM

fix $g \in G$

Let $\tau_g : X \to X$

$$\tau_g(x) := g * x$$

Then τ_g is a bijection

PROOF

$$(\tau_{g^{-1}} \circ \tau_g)(x)$$
$$= \tau_{g^{-1}}(\tau_g(x))$$
$$= \tau_{g^{-1}}(g * x)$$
$$= g^{-1} * (g * x)$$
$$= (g^{-1} \cdot g) * x$$
$$\overset{(ii)}{=} e * x$$
$$\overset{(i)}{=} x$$

$\tau_{g^{-1}} \circ \tau_g = id$

$\Rightarrow \tau_g \circ \tau_{g^{-1}} = id$

$\therefore \tau_g = (\tau_{g^{-1}})^{-1}$

so τ_g is a bijection.

(ii) Suppose that G acts on X.

Then $G \longrightarrow Perm(X)$

$$:= \{ f \mid f : X \to X \text{ s.t. } f \text{ is a bijection} \}$$

$$\phi : g \longrightarrow \tau_g$$

ϕ is a homomorphism

$$\phi(gh)(x) = \tau_{gh}(x)$$
$$= (gh) * x$$
$$\overset{(ii)}{=} g * (h * x) = g * (\tau_h(x) = \tau_g(\tau_h(x))$$
$$= (\tau_g \circ \tau_h)(x)$$
$$= (\phi(g) \circ \phi(h))(x)$$

$\forall x \in X$

$\Rightarrow \phi(gh) = \phi(g) \cdot \phi(h)$

So ϕ is a homomorphism.

Therefore, if G acts on X, then we get a homomorphism

$$\phi : G \longrightarrow Perm(X)$$

Conversely, let G be a group and $\phi : G \to Perm(X)$ be any homomorphism, then we can define an action of G on X:

$$g * x = \phi(g)(x)$$

(i) $e * x = \phi(e)(x) = id(x) = x$

↳ Homomorphism then $\phi(e) = id$

(ii) $(g.h) * x = \phi(g.h)(x) = (\phi(g) \circ \phi(h))(x)$

$$= \phi(g)(\phi(h)(x)) = \phi(g)(h * x) = g * (h * x)$$

$\square$

"An action of G on X is the same thing as a homomorphism $G \longrightarrow \mathrm{Perm}(X)$"

We say that the action of G on X is faithful if $\phi : G \longrightarrow \mathrm{Perm}(X)$ is injective

($\Leftrightarrow$ if $\phi(g) = id$ then $g = 1$
$\Leftrightarrow$ if $\phi(g)(x) = x, \forall x \in X$ then $g = 1$
$\Leftrightarrow$ if $g * x = x, \forall x \in X$ then $g = 1$)

We say that the action is transitive if given any $x, y \in X$ $\exists g \in G$, s.t $g * x = y$

"iki eleman alınan birini diğerine götüren bir eleman var mı?"

OBSERVATION

If $G \leq S_n = \mathrm{Perm}(\{1, 2, \dots\})$ then G acts on $\{1, 2, \dots, n\}$

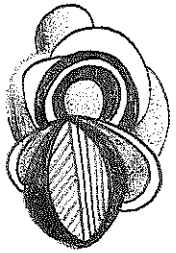
(i) $V = \{1, (12)(34), (13)(24), (14)(23)\} \leq S_4$

V acts on $\{1,2,3,4\}$ set

$(12)(34) * 1 = 2$.
$(12)(34) * 2 = 1$
$(12)(34) * 3 = 4$
$(12)(34) * 4 = 3$.

Look at each one

Is this action transitive?
Yes.



(ii) $\{1, (123), (132)\} \leq S_4$

$\{1, (123), (132)\}$ acts on $\{1, 2, 3, 4\}$

This action is not transitive.

(iii) Let $G = \mathbb{Z}$ and $X = \{a, b\}$

$\phi : \mathbb{Z} \longrightarrow \mathrm{Perm}(X)$

$\phi(z) = id, \forall z \in \mathbb{Z}$

This is not faithful.

Observation

Suppose that G acts on X. Then we have an equivalence relation on X:

$$x \sim y \Leftrightarrow \exists g \in G \text{ s.t } g * x = y$$

This is an equivalence relation. So this partitions X into equivalence classes $[x]$ for $x \in X$

$[x]$ is called the orbit of x under G

There is a single orbit in $X \Leftrightarrow$ the action of G on X is transitive.

▶ $G = \{1, (12), (34), (12)(34)\} \leq S_4$

then G acts on $\{1, 2, 3, 4\}$

$$\{1,2,3,4\} = \{1,2\} \dot\cup \{3,4\}$$
$$= [1] \dot\cup [3]$$

▶ Let $G = \langle \sigma \rangle \leq S_n$

then the equivalence classes in $\{1, 2, \dots, n\}$ under the action of G are the sets that appear in the disjoint cycle decomposition of σ.

▶ $G = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \;\middle|\; \begin{matrix} a, b, c, d \in \mathbb{R} \\ ad - bc \neq 0 \end{matrix} \right\} = GL_2(\mathbb{R})$

▶ $\mathcal{H} := \{ z \in \mathbb{C} \mid \operatorname{Im}(z) > 0 \}$

upper half plane

$G \times \mathcal{H} \longrightarrow \mathcal{H}$

$\begin{pmatrix} a & b \\ c & d \end{pmatrix} * z = \dfrac{az + b}{cz + d}$

Recall

$[x] = Gx$ New notation

NOTATION / DEFINATION

Given $x \in X$

$G_x = \{ g \in G \mid gx = x \}$

(stabilizer of x is G (= isotropy group of G)

PROPOSITION

$G_x \leq G$

PROOF

- $e \in G_x$, since $ex = x$
- if $g, h \in G_x$, then $gx = x$ and $hx = x$

Since $hx = x$

$x = ex$

$(h^{-1}h)x = h^{-1}(hx) = h^{-1}x$

$\Rightarrow h^{-1}x = x$

$(gh^{-1})x = g(h^{-1}x) = gx = x$

$h^{-1}x = x$

$\Rightarrow gh^{-1} \in G_x$

$\therefore G_x \leq G$

PROPOSITION

There is a natural 1-1, correspondence between

$\mathcal{L}_{G_x}(G)$ and Gx

COROLLARY

$(G : G_x) = |Gx|$

PROOF

$(G : G_x) = |\mathcal{L}_{G_x}(G)| = |Gx|$

PROOF

$$\mathcal{L}_{G_x}(G) = \{ g\, G_x \mid g \in G \} \underset{\alpha}{\overset{\beta}{\rightleftarrows}} Gx = \{ gx \mid g \in G \}$$

$\alpha(g \cdot G_x) = gx$

$\beta(g \cdot x) = g G_x$

Well-defined

- If $g G_x = g' G_x$ then $g^{-1} g' \in G_x \Rightarrow (g^{-1} g') \cdot x \Rightarrow g'x = gx$
  $\Rightarrow \alpha(g G_x) = \alpha(g' G_x)$
- If $gx = g'x$ then $g^{-1} g' x = x \Rightarrow g^{-1} g' \in G_x \Rightarrow g' G_x = g G_x$
  $\Rightarrow \beta(gx) = \beta(g'x)$
- $(\beta \circ \alpha)(g G_x) = \beta(\alpha(g \cdot G_x)) = \beta(gx) = g G_x \quad \therefore \beta \circ \alpha = id$
- $(\alpha \circ \beta)(gx) = \alpha(\beta(gx)) = \alpha(g \cdot G_x) = gx \quad \therefore \alpha \circ \beta = id$

So α and β are inverses of each other.

DEF Let $X_G = \{x \in X \mid gx = x, \forall g \in G\}$
$= \{x \in X \mid G_x = G\}$ $\longrightarrow$ $X_G = \{x \in X \mid Gx = \{x\}\}$

the set of fixed points of X under the G-action.

Corollary

If $|X|, |G| < \infty$ then $|[X]| = |Gx| \mid |G|$

Proof

$$|Gx| = (G : G_x) \mid |G|$$

Observation

Suppose $|X|, |G| < \infty$

$$X = \bigsqcup_{i \in I} Gx_i$$

$$= \bigsqcup_{i \in A} Gx_i \sqcup \bigsqcup_{i \in B} Gx_i = X_G \sqcup \bigsqcup_{i \in B} Gx_i$$

s.t. $|Gx_i| = 1$, for all $i \in A$ and $|Gx_i| > 1$ for all $i \in B$ such that $|Gx_i| > 1$

$$|X| = |X_G| + \sum_{i \in B} |Gx_i|$$

$$= |X_G| + \sum_{i \in B} \underbrace{(G : G_{x_i})}_{(G : G_{x_i}) > 1 \text{ and } (G : G_{x_i}) \mid |G|}$$

If $|G| = p^n$

$\Rightarrow p \mid (G : G_{x_i})$

$\bullet \Rightarrow |X| \equiv |X_G| \pmod p$

CAUCHY'S THEOREM

If $|G| < \infty$ and $p \mid |G|$ then G has an element of order p.

Corollary

If $|G| < \infty$ and $p \mid |G|$ then $\exists H \leq G$, s.t. $|H| = p$

"elementen ondersteunende subgroep"

Proof

$\exists g \in G$ s.t. $|\langle g \rangle| = |g| = p$

Take $H = \langle g \rangle$ $\square$

PROOF

Let

$$X = \{(g_1, \dots, g_p) \mid g_1 g_2 \cdots g_p = 1, g_i \in G\}$$

Let

$C = \langle (1\,2 \cdots p) \rangle \subseteq S_p$ $\qquad \alpha = (1 \cdots p)$
$= \langle \alpha \rangle$
$|C| = p$

Then C acts on X

$\alpha \in C$, $(g_1, g_2 \cdots g_p) \in X$

$\alpha \cdot (g_1, \dots, g_p) = g_{\alpha(1)} \cdots g_{\alpha(p)} \in X$

C acts on X and $|C| = p$

$\therefore |X| \equiv |X_C| \pmod p$ $\circledast$

$$X = \{(g_1, \dots, g_p) \mid g_1 \cdots g_p = 1\}$$
$$= \{(g_1, \dots, g_{p-1}, (g_1 \cdots g_{p-1})^{-1}) \mid g_1, \dots, g_p = 1\}$$

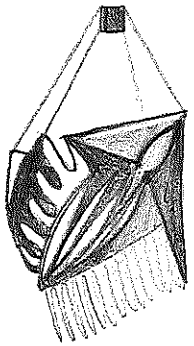
$|X| = |G|^{p-1}$ $\quad p \mid |G| \Rightarrow p \mid |X|$

$\circledast \Rightarrow p \mid |X_c|$

$$X_c = \{(g_1, \dots, g_p) \mid \forall \alpha \in C \quad (g_1, \dots g_p) = (g_{\alpha(1)} \cdots g_{\alpha(p)})\}$$

$$= \{(g, \dots, g) \mid g \cdots g = 1\}$$
$$= \{(g, \dots, g) \mid g^p = 1\}$$

$1 < |X_c|$, $\exists g \neq 1$ s.t. $(g, \dots, g) \in X_c$ then $|g| = p$



Recall

If $|G| = p^n$ acting on X, a finite set then

$|X| \equiv |X_G| \pmod p$

$X_G = \{x \in X \mid gx = x \ \forall g \in G\}$

- If G is a finite group and $p \mid |G|$ then G has an element of order p.

a conjugate of H

$H^g = \{h^g \mid h \in H\}$
$= \{g^{-1} h g \mid h \in H\}$

DEF: Let $H \leq G$, then the normalizer of H in G, denoted by

$$N_G(H) = \{g \in G \mid H^g = H\}$$

Let $X = \{H \mid H \leq G\}$ $\rightarrow$ the set of all subgroups of G.

$$G \times X \longrightarrow X$$
$$g \cdot H \longrightarrow H^{g^{-1}}$$

$$g_2(g_1 H) = g_2(H^{g_1^{-1}}) = (H g_1^{-1})^{g_2^{-1}} = H^{g_1^{-1} g_2^{-1}}$$
$$= H^{(g_1 g_2)^{-1}} = (g_2 g_1) H$$

Let $H \in X$

What is $G_H = \{g \in G \mid H^g = H\} = N_G(H) \leq G$.

$\uparrow$ isotropy group of H

$$(G : N_G(H)) = (G : G_H) = |\underbrace{G \cdot H}_{\text{orbit of } H \text{ under the action of } G}| = |\{H^g \mid g \in G\}| = \#\text{ of conjugates of } H$$

So $H \trianglelefteq G \Leftrightarrow N_G(H) = G$
$\Leftrightarrow$ # of conjugates of H is 1

PROPOSITION

Let $H \leq G$, s.t. $|H| = p^k$ then $(G:H) \equiv (N_G(H) : H) \pmod p$

PROOF

Let H act on $\mathcal{L}_H(G)$ as

$$H \times \mathcal{L}_H(G) \longrightarrow \mathcal{L}_H(G)$$
$$(h, gH) \longrightarrow hgH \qquad |H| = p^k$$

$\therefore$ By the lemma from the prev class

$$|\mathcal{L}_H(G)| \equiv |(\mathcal{L}_H(G))_H| \pmod p$$

$$(\mathcal{L}_H(G))_H = \{gH \mid hgH = gH \ \forall h \in H\}$$

$$= \{gH \mid g^{-1}hgH = H, \forall h \in H\}$$
$$= \{gH \mid g^{-1}hg \in H, \forall h \in H\}$$
$$= \{gH \mid g^{-1}Hg = H, \forall h \in H\}$$
$$= \{gH \mid g^{-1} \in N_G(H)\}$$
$$= \{gH \mid g \in N_G(H)\} = \mathcal{L}_H(N_G(H))$$

$$(G:H) = |\mathcal{L}_H(G)| \equiv |\mathcal{L}_H(N_G(H))| \pmod p$$
$$= (N_G(H):H)$$

Corollary

Let $H \leq G$, $|H| = p^n$ suppose that $p \mid (G:H)$

$\Rightarrow p \mid (N_G(H):H)$

SYLOW THEOREM (1)

Let G be a finite group of order $|G| = p^n m$ with $1 \leq n$ and $(m,p)=1$

1- G contains a subgroup of order p^i for every $1 \leq i \leq n$

2- If H is a subgroup of order p^i, w/ $1 \leq i < n$ then $\exists H' \leq G$ s.t. $|H'| = p^{i+1}$ and $H \trianglelefteq H'$.

PROOF

(1) We will do induction on n

$n=1$, follows by Cauchy

In general, if $n>1$,
use Cauchy's theorem to find a subgroup

$$T \leq G \text{ s.t. } |T| = p$$

$$0 \equiv \underbrace{(G:T)}_{p^{n-1}m} \equiv (N_G(T):T) \pmod p$$

since $n>1$

$$\Rightarrow p \mid (N_G(T):T)$$

But $T \trianglelefteq N_G(T)$

$$|N_G(T)/T| = p^\alpha m'$$
$$(m', p) = 1$$
$$1 \leq \alpha < n$$

$\Rightarrow N_G(T)$ has a subgroup of order p^i for every $1 \leq i \leq \alpha$.

By the induction hypothesis

$\Rightarrow G$ has a subgroup of order $1 \leq i \leq \alpha+1$

If $\alpha+1 = n$, then we are done, if not then we can continue to the argument by taking T a subgroup of order $\alpha+1 < n$

$T \trianglelefteq N_G(T)$ and $p \mid (N_G(T):T)$

$\Rightarrow H' \trianglelefteq N_G(\mathcal{T})/\mathcal{T}$

Cauchy's Theorem

$|H'| = p$

$\mathcal{T} \leq \pi^{-1}(H') \leq G$

$|\pi^{-1}(H')| = p^{(\alpha+1)+1}$

Remember one to one correspondence

$\pi : N_G(\mathcal{T}) \longrightarrow N_G(\mathcal{T})/\mathcal{T}$

(2) $H \leq G$

$|H| = p^i$

$p \mid (G:H)$

$\Rightarrow p \mid (N_G(H):H)$

$\Rightarrow p \mid |N_G(H)/H|$

$\Rightarrow \mathcal{T} \leq N_G(H)/H$

Cauchy $|\mathcal{T}| = p$

$\pi : N_G(H) \longrightarrow N_G(H)/H$

$\vee\!/$

$\mathcal{T}$

$H \leq \pi^{-1}(\mathcal{T}) \leq N_G(H)$

$\underbrace{\pi^{-1}(\mathcal{T})/H}_{\text{order } p} \xrightarrow{\sim} \mathcal{T}$ has order p

$|H'| = |\pi^{-1}(\mathcal{T})| = |\pi^{-1}(\mathcal{T})/H| \; |H|$

$= p \cdot p^i = p^{i+1}$

since $\boxed{H \trianglelefteq N_G(H)}$ and

$H' \leq N_G(H) \Rightarrow H \trianglelefteq H'$

DEF With notation as above if $H \leq G$ w/ $|H| = p^n$

Then H is called a Sylow p-subgroup of G.

$|X| \equiv |X_G| \pmod p$

SYLOW THEOREM (2)

If P_1 and P_2 are two sylow p-subgroups of G Then P_1 and P_2 are conjugate.

<u>PROOF</u> Let P_1 act on $\mathcal{L}_{P_2}(G)$

$$P_1 \times \mathcal{L}_{P_2}(G) \longrightarrow \mathcal{L}_{P_2}(G)$$
$$(\alpha, gP_2) \longrightarrow \alpha g P_2$$

$|P_1| = p^n$ so

$\underset{\substack{\| \\ m}}{(G:P_2)} = |\mathcal{L}_{P_2}(G)| \equiv |(\mathcal{L}_{P_2}(G))_{P_1}| \pmod p$

$p \nmid m \Rightarrow p \nmid |(\mathcal{L}_{P_2}(G))_{P_1}|$

$\Rightarrow (\mathcal{L}_{P_2}(G))_{P_1} \neq \emptyset$

$(\mathcal{L}_{P_2}(G))_{P_1} = \{gP_2 \mid p_1 g P_2 = gP_2, \forall p_1 \in P_1\}$

$= \{gP_2 \mid g^{-1} p_1 g P_2 = P_2, \forall p_1 \in P_1\}$

$= \{gP_2 \mid g^{-1} p_1 g \in P_2, \forall p_1 \in P_1\}$

$= \{gP_2 \mid g^{-1} P_1 g \subseteq P_2\} = \{gP_2 \mid g^{-1} P_1 g = P_2\}$

$\Rightarrow \exists g \in G$ s.t. $g^{-1} P_1 g = P_2$

Observation

Let $\mathcal{S}_p$ denote the set of all sylow p-subgroups of G

$$\mathcal{S}_p = \{H \mid H \leq G, |H| = p^n\}$$

G acts on $\mathcal{S}_p$ by conjugation

$$*: G \times \mathcal{S}_p \to \mathcal{S}_p$$
$$(g, H) \longrightarrow H^{g^{-1}} \quad \text{this is an action}$$
$$|H^{g^{-1}}| = |H| = p^n$$

Let $P \in \mathcal{S}_p$ and $Q \in \mathcal{S}_p$

$$\Longrightarrow \quad P^{g^{-1}} = Q$$

$\exists g \in G$

$\therefore$ The orbit of P under the action $(*)$ is all of $\mathcal{S}_p$.

i.e

$$G * P = \{P^{g^{-1}} \mid g \in G\} = \mathcal{S}_p$$

$$(G : G_P) = |G * P| = |\mathcal{S}_p| = n_p$$

n_p = # of Sylow p-subgroup

isotrophy

Corollary $\quad n_p = [G : N_G(P)] \mid |G|$

SYLOW THEOREM (3) If G is a finite group and $p \mid |G|$ then $n_p \equiv 1 \pmod p$ and $n_p = [G : N_G(P)] \mid |G|$

PROOF

Let $\mathcal{P} \in \mathcal{S}_p$ and let

$\mathcal{P}$ act on $\mathcal{S}_p$ by conjugation

$$\mathcal{P} \times \mathcal{S}_p \longrightarrow \mathcal{S}_p$$
$$(g, H) \longrightarrow H^{g^{-1}}$$

$$|\mathcal{P}| = p^n$$

$$\Rightarrow |\mathcal{S}_p| \equiv |(\mathcal{S}_p)_{\mathcal{P}}| \pmod p$$

$$(\mathcal{S}_p)_{\mathcal{P}} = \{H \in \mathcal{S}_p \mid \forall g \in \mathcal{P}, H^{g^{-1}} = H\}$$
$$= \{H \mid H \leq G, |H| = p^n, H^{g^{-1}} = H, \forall g \in \mathcal{P}\}$$
$$= \{H \mid H \leq G, |H| = p^n, \mathcal{P} \leq N_G(H)\}$$

$$H \leq N_G(H) \leq G$$

$p^n = |H| \quad |N_G(H)| = p^n m' \quad p^n m = |G|$

$(p, m') = 1 \quad (p, m) = 1$

So H is a sylow p-subgroup of $N_G(H)$

But $|\mathcal{P}| = |H| = p^n$

So since

$$\mathcal{P} \leq N_G(H)$$

$\mathcal{P}$ is also a sylow p-subgroup of $N_G(H)$

By Sylow theorem (2) applied to $\mathcal{P}$, H and $N_G(H)$ $\mathcal{P}$ and H are conjugate in $N_G(H)$, i.e. $\exists g \in N_G(H)$ s.t.

$$H = H^g = P$$
$$g \in N_G(H)$$

THEOREM If G is a finite p-group, i.e. $|G| = p^n$, then

$$Z(G) \neq 1$$

PROOF

Let G act on G by conjugation

$$G \times G^{(=X)} \longrightarrow G^{(=X)}$$
$$(g, \alpha) \longrightarrow \alpha^{g^{-1}}$$

Since G is a p-group

$$0 \equiv p^n = |G| \equiv |G_G^{(=X)}| \pmod p$$

$$\begin{aligned} G_G &= \{\alpha \in G \mid \alpha^{g^{-1}} = \alpha, \forall g \in G\} \\ &= \{\alpha \in G \mid g\alpha g^{-1} = \alpha \ \forall g \in G\} \\ &= \{\alpha \in G \mid g\alpha = \alpha g \ \forall g \in G\} \\ &= Z(G) \end{aligned}$$

$|Z(G)| \equiv 0 \pmod p$

$\Rightarrow p \mid |Z(G)|$

$\Rightarrow \{1\} \lneq Z(G)$ $\square$

Lemma

Suppose $H, K \trianglelefteq G$ s.t. $H \cap K = 1$, and

$$HK = \{hk \mid h \in H, k \in K\} = G$$

Bir normaise subgroup olmayor.

then

$$G \xleftarrow{\sim} H \times K$$

PROOF

Let $h \in H$, $k \in K$

$\underbrace{hkh^{-1}}_{\in K}k^{-1} \in K$

(since $K \trianglelefteq G$)

$\underbrace{hkh^{-1}}_{\in H}k^{-1} \in H$

$\therefore hkh^{-1}k^{-1} \in H \cap K = \{1\}$

$\Rightarrow hk = kh$

Define

$$\varphi : H \times K \longrightarrow G$$
$$\varphi(h,k) = hk$$

$\varphi((h_1k_1)(h_2k_2))$

$= \varphi(h_1h_2, k_1k_2) = h_1h_2k_1k_2 = h_1k_1h_2k_2 = \varphi(h_1k_1)\varphi(h_2k_2)$

$\therefore$ Homomorphism.

$\ker\varphi := \{(h,k) \mid \varphi(h,k) = 1\}$

$= \{(h,k) \mid hk = 1\}$

$= \{(h,k) \mid \underset{H}{h} = \underset{K}{k^{-1}}\}$

$= \{(1,1)\}$

$\therefore \varphi$ is injective

Since $HK = G$ $\therefore \varphi$ is surjective

$\therefore \varphi$ is isomorphism

THEOREM

If $|G|=p^2$ then G is abelian $\left(\begin{array}{l} \text{FTFGAP} \\ \Rightarrow G \simeq \mathbb{Z}/p \times \mathbb{Z}/p \\ G \simeq \mathbb{Z}/p^2 \end{array} \right)$

PROOF

Suppose G is not cyclic (b/c if G is cyclic then $G \simeq \mathbb{Z}/p^2$ so is abelian)

Then if $g \in G \setminus \{1\}$ then $|g|=p$

Take $\alpha \in G \setminus \{1\}$, $|\langle \alpha \rangle|=p$

take $\beta \in G \setminus \langle \alpha \rangle$, $|\langle \beta \rangle|=p$

$H=\langle \alpha \rangle \trianglelefteq G$ (By Sylow 1)

$K=\langle \beta \rangle \trianglelefteq G$ (By Sylow 1)

$H \cap K = \{1\}$ ($\mathbb{Z}_p \times \mathbb{Z}_p$ formatunda)

▶ If $|G|=15$ then G is cyclic
$=3\cdot 5$

Let P_3 be a Sylow 3-subgroup of G

P_5 " " 5 " "

$n_3 \mid 15$ $\quad n_3 \equiv 1 \pmod 3 \Rightarrow n_3 \mid 5 \Rightarrow n_3 = 1$

$n_5 \mid 15$ $\quad n_5 \equiv 1 \pmod 5 \Rightarrow n_5 \mid 3 \Rightarrow n_5 = 1$

This implies that

$\Rightarrow P_3 \trianglelefteq G$ → Then G is not simple.

$n_3=1$ ise $N_G(P_3)=G$, $P_3 \trianglelefteq G$

$P_5 \trianglelefteq G$

$\boxed{\begin{array}{l} P_3 \cap P_5 \leq P_3 \\ \leq P_5 \end{array}}$

$|P_3 \cap P_5| \mid 3$ and $|P_3 \cap P_5| \mid 5$

$\Rightarrow P_3 \cap P_5 = 1$

$P_3 \leq P_3 P_5 \leq G$

$P_5 \leq P_3 P_5$

$3 \mid |P_3 P_5|$ $\quad |P_3 P_5| \mid 15$

$5 \mid |P_3 P_5|$ $\quad$ [illegible] $\Rightarrow P_3 P_5$ [illegible]

$G \simeq P_3 \times P_5 \simeq \mathbb{Z}_3 \times \mathbb{Z}_5$
$\simeq \mathbb{Z}_{15}$

Proposition

$H \leq G$, $N \trianglelefteq G$

$HN \leq G$

PROOF

Let $h_1 \cdot n_1 h_2 n_2$

$= \underbrace{h_1 h_2}_{\in H} \underbrace{h_2^{-1} n_1 h_2}_{\in N} \underbrace{n_2}_{\in N}$

$(h \cdot n)^{-1} = n^{-1} h^{-1}$

$= h^{-1} \underbrace{h n^{-1} h^{-1}}_{\in N} \in HN$

$(N \trianglelefteq G)$

▶ Do it for P_3 and P_7

$n_3 \mid 21$ $\quad n_3 \equiv 1 \pmod 3$ $\quad n_3 \mid 7$ → 7 de olabilir b/c. oyeden olmaz

$n_7 \mid 21$ $\quad n_7 \equiv 1 \pmod 7$

▶ Find the Sylow 2 and 3 subgroup of S_3

$|S_3|=6$

$\mathcal{S}_2 = \{\{1,(12)\},\{1,(13)\},\{1,(23)\}\}$

$\mathcal{S}_3 = \{\{1,(123),(132)\}\}$

$n_2=3$ ($n_2 \mid 6$ $\quad n_2 \equiv 1 \pmod 2$)

$n_3=1$ ($n_3 \mid 6$ $\quad n_3 \equiv 1 \pmod 3$)

▶ If $|G|=p\cdot q$ with $p<q$

then if P_q is a Sylow q-subgroup then

$P_q \trianglelefteq G$

$n_q \mid p\cdot q$, $n_q \equiv 1 \pmod q$ $\Rightarrow n_q \mid p$ $\quad \boxed{n_q=p \text{ olamaz çünkü } p<q}$

$\Rightarrow n_q = 1$

FUNDAMENTAL THEOREM OF FINITELY GENERATED ABELIAN GROUPS

Proposition

Suppose that A is an abelian group and $X \subseteq A$
Then the following are equivalent

(i) $\forall a \in A$, there exists unique $x_1, \dots, x_n \in X$ (pairwise distinct) and $k_1, \dots, k_n \in \mathbb{Z}$ s.t.
$a = k_1 x_1 + \dots k_n x_n$

(ii) X generates A and if $k_1 x_1 + \dots + k_n x_n = 0$ w/ $x_i \in X$ (and $x_i \neq x_j$ for $i \neq j$) and $k_1, \dots, k_n \in \mathbb{Z}$ then $k_1 = k_2 = \dots = k_n = 0$

(i) $\Rightarrow$ (ii) : o.k.

(ii) $\Rightarrow$ (i) Let $a \in A$. since X generates A,

$$a = k_1 x_1 + \dots + k_n x_n \quad \text{for some } k_1, k_2 \dots k_n \in \mathbb{Z}$$

Suppose $a = k_1' x_1 + \dots k_n' x_n$

$0 = (k_1' - k_1) x_1 + \dots + \dots (k_n' - k_n) x_n$

$\Rightarrow k_i' - k_i = 0$, for all $1 \leq i \leq n$

$\Rightarrow k_i' = k_i$, for all $1 \leq i \leq n$

DEF: We say A is a <u>free abelian</u> group w/ basis X if $X \subseteq A$ satisfies the hypothesis above

▶ $\mathbb{Z} \oplus \mathbb{Z}$ is free abelian group with basis $\{(1,0), (0,1)\}$

▶ $\mathbb{Z}_6$ is not a free abelian group (with any basis)
▷ $6x = 0$ $\forall x \in \mathbb{Z}_6$ and $n \neq 0$

DEF A group G is called <u>torsion-free</u> if

$$G_{tors} = \{ g \in G \mid |g| < \infty \} = \{1\}$$

PROPOSITION

If A is a free abelian group with basis X, then A is torsion-free

PROOF

Suppose $a \in A$ <u>has finite order</u>, $m \in \mathbb{N}$ and write

$a = k_1 x_1 + \dots + k_n x_n$

$0 = m.a = (mk_1) x_1 + \dots + (mk_n) x_n$

$\Rightarrow m k_i = 0$, for all i

$\Rightarrow k_i = 0$, for all i

$\Rightarrow \underline{a = 0}$

HW

- $(\mathbb{Q}, +)$ is torsion free but not free abelian
- Prove that if A is <u>abelian</u> and <u>f.g</u> and <u>torsion free</u> then A is free abelian

FACT

If A is a free abelian group w/ basis X and also a free abelian group w/ basis Y then

$$|X| = |Y|$$

We will prove this when $|X| < \infty$

OBSERVATION

Suppose $|X| < \infty$ and A is a free abelian group w/ basis X, then

$$A \simeq \underbrace{\mathbb{Z} \oplus \mathbb{Z} \oplus \dots \oplus \mathbb{Z}}_{n \text{ times}} \quad \text{where } |X| = n$$

suppose $X = \{x_1, \dots, x_n\}$

$$\mathbb{Z} \oplus \mathbb{Z} \oplus \dots \mathbb{Z} \xrightarrow{\phi} A$$
$$(k_1, \dots, k_n) \longrightarrow k_1 x_1 + \dots + k_n x_n$$

$$\mathbb{Z} \oplus \mathbb{Z} \oplus \dots \oplus \mathbb{Z} \xleftarrow{\psi} A$$
$$(k_1, \dots, k_n) \qquad \theta = k_1 x_1 + \dots k_n x_n$$

$$\psi \circ \phi = id = \phi \circ \psi$$

suppose that A is any abelian group and $m \in \mathbb{N}$

$$mA = \{m \cdot a \mid a \in A\} \leq A$$

$$(m \cdot a + m \cdot b = m(a+b) \in mA)$$

A is abelian

A/mA makes sense.

If A is a free abelian group w/ basis X, then $A \simeq \mathbb{Z} \oplus \mathbb{Z} \oplus \dots \mathbb{Z}$

$$A/mA \simeq (\mathbb{Z} \oplus \dots \oplus \mathbb{Z}) / m(\mathbb{Z} \oplus \dots \oplus \mathbb{Z})$$

$$(\mathbb{Z} \oplus \dots \oplus \mathbb{Z}) / m(\mathbb{Z} \oplus \dots \oplus \mathbb{Z})$$

$$\simeq (\mathbb{Z} \oplus \dots \oplus \mathbb{Z}) / \{(mk_1, \dots, mk_n) \mid k_i \in \mathbb{Z}\}$$

$$\simeq \underbrace{\mathbb{Z}_m \oplus \dots \oplus \mathbb{Z}_m}_{n \text{ times}}$$

$$|A/mA| = m^n = m^{|X|}$$

If Y is another basis then

$$|A/mA| = m^{|Y|}$$

$$\Rightarrow |X| = |Y|$$

DEF: If A is a free abelian group w/ basis X. We call the cardinality $|X|$ of X, the rank of A

("some" people, namely Lang, calls it Betti number)

THE FILE

Lemma

suppose that $\{x_1, \dots, x_n\}$ is a basis for A, let $i \neq j$ and $t \in \mathbb{Z}$. then

$$X' = \{x_1, \dots, x_{i-1}, x_i + x_j, x_{i+1}, \dots, x_j, \dots x_n\}$$

is also a basis for A.

PROOF

Need to show that X' generates A

Let $a \in A$, since X generates A

$\exists k_1, \dots, k_n \in \mathbb{Z}$ s.t.

$$a = k_1 x_1 + \dots + k_n x_n$$
$$= k_1 x_1 + \dots k_i x_i + \dots k_n x_n$$
$$= k_1 x_1 + \dots + k_i (x_i + t x_j) + \dots + (k_j + t k_i) x_j + \dots + k_n \underbrace{x_n}_{\in X'}$$

UNIQUENESS

suppose

$$0 = k_1 x_1 + \dots + k_{i-1} x_{i-1} + k_i (x_i + t x_j) + \dots + k_n x_n$$
$$= k_1 x_1 + \dots + k_{i-1} x_{i-1} + k_i x_i + \dots + (k_i t + k_j) x_j + \dots$$

$\Longrightarrow$ $k_1 = 0, \dots k_{i-1} = 0, k_i = 0$

X is a basis $\qquad k_i t + k_j = 0 \quad \dots k_n = 0$

$$\Longrightarrow k_1 = k_2 = \dots = k_n = 0$$

THEOREM If A is a finitely generated abelian group then

$$A \simeq \mathbb{Z}_{d_1} \oplus \mathbb{Z}_{d_2} \oplus \cdots \oplus \mathbb{Z}_{d_k} \oplus \mathbb{Z} \oplus \mathbb{Z}$$

s.t. $d_1 | d_2, d_2 | d_3 | \cdots, d_{k-1} | d_k$
and $d_1, \dots d_k$ are unique and also n is unique,

(n is called the rank of the Betti number of A)

It sufficies to prove the following Lemma

Lemma: If $F \leq \underbrace{\mathbb{Z} \oplus \mathbb{Z} \oplus \cdots \oplus \mathbb{Z}}_{m\text{-times}} = A$

then there exists a basis

$x_1, \dots, x_m$ of $\mathbb{Z} \oplus \mathbb{Z} \oplus \cdots + \mathbb{Z}$

and $d_1, \dots, d_k$ s.t $d_1 | d_2, \dots, d_{k-1} | d_k$
and $\{d_1 x_1, \dots, d_k x_k\}$ is a basis for F.

In particular, F is a free abelian group.

$$\Rightarrow (\mathbb{Z} \oplus \cdots \oplus \mathbb{Z}) / F$$

$$\simeq \mathbb{Z} \oplus \cdots \oplus \mathbb{Z} / (d_1 a_1, d_2 a_2, \dots, d_k a_k, 0, 0 \dots 0)$$

$$\simeq \mathbb{Z}_{d_1} \oplus \mathbb{Z}_{d_2} \oplus \cdots \oplus \mathbb{Z}_{d_k} \oplus \mathbb{Z} \oplus \mathbb{Z} \oplus \cdots \oplus \mathbb{Z}$$

Consider all possible bases for A, $\{y_i\}_{1 \leq i \leq n}$

and all possible combinations

$$a_1 y_1 + \cdots + a_m y_m \in F$$

and among these let $|a_1|$ be the smallest positive integer

$$|a_1| = d_1$$

then

$$d_1 y_1 + a_2 y_2 + \cdots + a_m y_m \in F$$

Claim $d_1 | a_i$, for every y

PROOF

$$a_i = d_i q_i + r_i$$

$$d_1 y_1 + a_2 y_2 + \cdots + a_m y_m \in F$$

$$d_1 y_1 + (d_1 q_2 + r_2) y_2 + \cdots + a_m y_m \in F$$

$$d_1 \underbrace{(y_1 + q_2 y_2)}_{y_1'} + r_2 y_2 + \cdots + a_m y_m$$

$\{y_1', y_2, \dots, y_n\}$ is another basis for A

$$\underset{r_2 < d_1}{\Longrightarrow} r_2 = 0 \Longrightarrow d_1 | a_2$$

$d_1 | a_i$, for every $2 \leq i \leq m$

Consider now all bases.

$$\{y_1, z_2, \dots, z_m\}$$

s.t. $\exists\ d_1 y_1 + a_2 z_2 + \cdots + a_m z_m \in F$

(NOTE THAT This implies $d_1 | a_i$, for all i)

Among all these expressions let $|a_2| = d_2$ be the smallest one possible then $d_1 | d_2$

CLAIM

If

$$d_1y_1 + d_2y_2 + a_3z_3 + \dots + \qquad (d_2y_2 = d_2q_3 + r_3)$$

then $d_2 \mid a_i$ for all $3 \le i \le m$, $0 \le r_3 < d_2$

$$d_1y_1 + d_2(y_2 + q_3z_3) + r_3z_3 + \dots + a_m z_m \in F \qquad (y_2 + q_3z_3 = y_2')$$

$\{y_1, y_2', z_3, \dots, z_m\}$ is another basis

for $F \xRightarrow[r_3<d_2]{} r_3 = 0$

$\Longrightarrow d_2 \mid a_i$, for all i

By induction, we arrive at the desired basis.

Corollary If B is a f.g. abelian group then

$$B \xrightarrow{\sim} \mathbb{Z}_{d_1} \oplus \mathbb{Z}_{d_2} \oplus \dots \oplus \mathbb{Z} \dots \oplus \mathbb{Z}$$

s.t. $d_i \mid d_{i+1}$

Recall If $n = p_1^{s_1} \dots p_r^{s_r}$

$$\mathbb{Z}_n \simeq \mathbb{Z}_{p_1^{s_1}} \oplus \dots \oplus \mathbb{Z}_{p_r^{s_r}}$$

Corollary

Then if B is a finitely generated abelian group

$$B \simeq \mathbb{Z}_{p_1^{t_1}} \oplus \dots \oplus \mathbb{Z}_{p_\ell^{t_\ell}} \oplus \mathbb{Z} \oplus \dots \oplus \mathbb{Z}$$

Uniqueness

If

$$\mathbb{Z}_{p_1^{t_1}} \oplus \dots \oplus \mathbb{Z}_{p_\ell^{t_\ell}} \oplus \underbrace{\mathbb{Z} \oplus \dots \oplus \mathbb{Z}}_{a \text{ copies}}$$

$$\simeq \mathbb{Z}_{q_1^{s_1}} \oplus \dots \oplus \mathbb{Z}_{q_m^{s_m}} \oplus \underbrace{\mathbb{Z} \oplus \dots \oplus \mathbb{Z}}_{b \text{ copies}}$$

where $p_1 \le p_2 \le \dots \le p_\ell$

$q_1 \le \dots \le q_m$

and if $p_i = p_{i+1}$ then $t_i \le t_{i+1}$

if $q_i = q_{i+1}$ then $s_i \le s_{i+1}$

then $a = b$,

$p_i = q_i$ and $t_i = s_i$

PROOF

$$A_{tors} = \{a \in A \mid |a| < \infty\} \le A$$

True if A is abelian

$$B_{1,tors} \simeq \mathbb{Z}_{p_1^{t_1}} \oplus \dots \oplus \mathbb{Z}_{p_\ell^{t_\ell}}$$

$$B_{2,tors} \simeq \mathbb{Z}_{q_1^{s_1}} \oplus \dots \oplus \mathbb{Z}_{q_m^{s_m}}$$

$$B_{1,tors} \simeq B_{2,tors}$$

$$\mathbb{Z}_{p_1^{t_1}} \oplus \dots \oplus \mathbb{Z}_{p_\ell^{t_\ell}} \xrightarrow{\sim} \mathbb{Z}_{q_1^{s_1}} \oplus \dots \oplus \mathbb{Z}_{q_m^{s_m}}$$

$$B_1 / B_{1,tors} \simeq B_2 / B_{2,tors}$$

$$\| \qquad\qquad \|$$

$$\underbrace{\mathbb{Z} \oplus \dots \oplus \mathbb{Z}}_{a \text{ copies}} \qquad \underbrace{\mathbb{Z} \oplus \dots \oplus \mathbb{Z}}_{b \text{ copies}}$$

Then $a = b$

WLOG
$a=b=0$

$$A_1 \simeq A_2$$

$$A_1 = \mathbb{Z}_{p_1^{t_1}} \oplus \cdots \oplus \mathbb{Z}_{p_l^{t_l}} \qquad A_2 = \mathbb{Z}_{q_1^{s_1}} \oplus \cdots \oplus \mathbb{Z}_{q_m^{s_m}}$$

$$\Longrightarrow A_1(p_1) \simeq A_2(p_1)$$

$$A_1(p_1) \supseteq \mathbb{Z}_{p_1^{t_1}} \oplus \cdots \oplus \mathbb{Z}_{p_1^{t_g}} \qquad A_2(p_1) \supseteq \mathbb{Z}_{p_1^{s_1}} \oplus \cdots \oplus \mathbb{Z}_{p_1^{s_h}}$$

$t_1 \leq \cdots \leq t_g$ $\qquad$ $s_1 \leq \cdots \leq s_h$

$p_1^{t_g}$ is the order of the largest cyclic subgroup in $A_1(p_1)$

$p_1^{s_h}$ is the order of the largest cyclic subgroup in $A_2(p_1)$

$$\Longrightarrow p_1^{t_g} = p_1^{s_h}$$

Let $c_i \leq A_i(p_1)$ s.t. c_i is cyclic and $|c_i| = p_1^{t_g} = p_1^{s_h}$

$$A_1(p_1)/c_1 \simeq \mathbb{Z}_{p_1^{t_1}} \oplus \cdots \oplus \mathbb{Z}_{p_1^{t_{g-1}}}$$

$$\wr\!\wr$$

$$A_2(p_1)/c_2 \simeq \mathbb{Z}_{p_1^{s_1}} \oplus \cdots \oplus \mathbb{Z}_{p_1^{s_{m-1}}}$$

By induction

THEOREM If $\mathbb{Z}_{d_1} \oplus \cdots \oplus \mathbb{Z}_{d_m} \oplus \underbrace{\mathbb{Z} \oplus \cdots \oplus \mathbb{Z}}_{a \text{ copies}}$

$$\simeq \mathbb{Z}_{e_1} \oplus \cdots \oplus \mathbb{Z}_{e_n} \oplus \underbrace{\mathbb{Z} \oplus \cdots \oplus \mathbb{Z}}_{b \text{ copies}}$$

w/ $d_i \mid d_{i+1}$, $e_i \mid e_{i+1}$, for all i.

Then $a=b$ and $d_i = e_i$ for all i.

RINGS AND FIELDS

RING: A ring $(R,+,\cdot)$ is a non-empty set R, together with two operations, $+,\cdot$ on R, such that

(i) $(R,+)$ is an abelian group

(ii) $\forall a,b,c \in R$, $(a\cdot b)\cdot c = a(b\cdot c)$ Associativity for $\bullet$

(iii) Distributive Law $\forall a,b,c \in R$, $(a+b)c = a\cdot c + bc$ and $a(b+c) = ab+ac$

If there exists a $1 \in R$ s.t. $\forall a \in R$ $1a = a = a1$
Then we say that R is a ring with unity

If for every $a,b \in R$,
$ab = b\cdot a$
Then we say that R is commutative.

For us the most important type of rings will be commutative rings with unity

▶ $(\mathbb{Z}, +, \cdot)$ commutative ring with unity

• $(M_{22}(\mathbb{R}), +, \cdot)$

2x2 matrices with entries from $\mathbb{R}$

A ring with unity, but it is non-commutative

$$\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\neq$$

$$\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

• $LM(V,V)$ – not commutative ring w/unity.

• $(\mathbb{Z}_n, +, \cdot)$ commutative ring w/unity

Observation

(i) Note

$$0.r = (0+0).r = 0.r + 0.r$$

$$0 = (-0.r) + 0.r = -(0.r) + (0.r + 0.r)$$
$$= (-0.r + 0.r) + 0.r$$
$$= 0 + 0.r = 0.r$$

similarly, $r.0 = 0$, $\forall r \in R$

(ii) Suppose R has a unity 1

$1_R = 0 \Leftrightarrow R = \{0\}$ called trivial ring

$(\Rightarrow r = r.1 = r.0 = 0, R = \{0\})$
$\Leftarrow$ Trivial

(iii) Suppose that R is a comm. ring with unity, then we let

$$R^\times = \{r \in R \mid \exists s \in R, \text{ w/ } r.s = 1\}$$

called the invertible elements
– called the units

▨ $(R^\times, \cdot)$ is a group

• $r_1, r_2 \in R^\times$
then $\exists s_1, s_2 \in R^\times$ s.t.
$r_i s_i = 1$
$r_1 . r_2 s_2 s_1 = r_1 . 1 . s_1 = r_1 s_1 = 1$
$\Rightarrow r_1 r_2 \in R^\times$

$(R^\times, \cdot)$ is closed

- Assoc
- $1 \in R^\times$
- If $r \in R^\times$, then $\exists s \in R$ w/ $r.s = 1$, $\Rightarrow s \in R^\times$

So every element has an inverse

$\therefore (R^\times, \cdot)$ is a group

It is comm. since $(R, \cdot)$ is commutative.

▶ – $(\mathbb{Z}, +, \cdot)$, $\mathbb{Z}^\times = \{\pm 1\}$

– $(\mathbb{R}, +, \cdot)$, $\mathbb{R}^\times = \mathbb{R} \setminus \{0\}$

– $(\mathbb{Z}_n, +, \cdot)$, $\mathbb{Z}_n^\times = \{m \mid 0 \leq m < n, (m,n) = 1\}$

$|\mathbb{Z}_n^\times| = \varphi(n) \Rightarrow \forall 0 \leq a < n$,
$a^{\varphi(n)} \equiv 1 \pmod n$

EULER'S THEOREM

DEF We say that R is an <u>integral domain</u>, if R is a commutative ring w/ unity, s.t. $R \neq \{0\}$ and $\forall r, s \in R$,
$r.s = 0 \Rightarrow r = 0$ or $s = 0$

▶ • $(\mathbb{Z}, +, \cdot)$ integral domain

• $(\mathbb{Z}_n, +, \cdot)$ is not an integral domain
$\Leftrightarrow n$ is composite

$(a.b = 0$ in $\mathbb{Z}_n \Leftrightarrow n \mid a.b)$

$2.3 = 0$ in $\mathbb{Z}_6$

• $(M_{2\times 2}, +, \cdot)$

$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

• $R = \{a + b\alpha, a, b \in \mathbb{R}\}$
$(a + b\alpha)(c + d\alpha) = ac + (bc + ad)\alpha$
$(a + b\alpha) + (c + d\alpha) = (a+c) + (b+d)\alpha$
$\alpha . \alpha = 0$

$(\mathbb{R}, +, \cdot)$ comm w/ unity but

DEF

We say that $(F,+,\cdot)$ is <u>a field</u> if $F \neq \{0\}$ and $(F,+,\cdot)$ is a comm. ring w/ unity s.t.

$$F^\times = F \setminus \{0\}$$

▶ $(\mathbb{Z},+,\cdot)$ not a field

$(\mathbb{Q},+,\cdot)$, $\mathbb{Q}^\times = \mathbb{Q}\setminus\{0\}$ Field

$(\mathbb{R},+,\cdot)$, $\mathbb{R}^\times = \mathbb{R}\setminus\{0\}$ Field

$\mathbb{C}^\times = \mathbb{C}\setminus\{0\}$ Field

Remark

If F is a field then F is an integral domain

If $r.s=0$ in F and $r\neq 0 \Rightarrow r \in \underline{F\setminus\{0\}} = F^\times$

$\Rightarrow \exists r' \in F$ s.t. $r'.r = 1$

$\Rightarrow 0 = r'.0 = r'.r.s = 1.s = s$

▶ $(\mathbb{Z}_p,+,\cdot)$ is a field

If $r \in \mathbb{Z}_p \setminus \{0\}$

$\Rightarrow (r,p)=1$

$\Rightarrow \exists x,y \in \mathbb{Z}$ s.t.

$r.x + p.y = 1$

$r.x = 1 \pmod p$

$\Rightarrow r \in \mathbb{Z}_p^\times$

$(\mathbb{Z}_p,+,\cdot)$ is a field.

NOTE: We will show that there is a field of order p^n, for every prime power p^n.

Warning This is not

$(\mathbb{Z}_{p^n},+,\cdot)$.

Not a field.

$p.p^{n-1} = 0$ in $\mathbb{Z}_{p^n}$.

HOMOMORPHISM

Let R and S be two rings

A homomorphism φ from R to S is a function

$$\varphi : R \to S$$

s.t.

(i) $\varphi(r_1+r_2) = \varphi(r_1)+\varphi(r_2)$ $(\Rightarrow \varphi(0)=0)$

(ii) $\varphi(r_1.r_2) = \varphi(r_1).\varphi(r_2)$

(iii) $\varphi(1)=1$

$\ker\varphi = \{r\in R \mid \varphi(r)=0\}$

$\varphi : \mathbb{Z} \to \mathbb{Z}_n$

$a \to a \pmod n$

$\ker\varphi = n\mathbb{Z}$

DEF Let R be a ring, <u>an ideal</u> $\emptyset \neq I \subseteq R$ is a subset satisfying

(i) $(I,+)$ is a subgroup of $(R,+)$

(ii) If $\alpha \in I$ and $r \in R$

$\alpha.r \in I$

PROPOSITION

If $\varphi : R \to S$ is a homomorphism then $\ker\varphi \subseteq R$ is an ideal

PROOF

Recall $\ker\varphi = \{\alpha\in R \mid \varphi(\alpha)=0\}$

- If $r\in R$, $\alpha \in \ker\varphi$ then

$\varphi(r.\alpha) = \varphi(r)\varphi(\alpha)$
$= \varphi(r).0 = 0$

$\Rightarrow r\alpha \in \ker\varphi$

- If $\alpha,\beta \in \ker\varphi$ then $\varphi(\alpha+\beta)$
$= \varphi(\alpha)+\varphi(\beta)$
$= 0+0 = 0$

$\Rightarrow \alpha+\beta \in \ker\varphi$

If $\alpha \in \ker\varphi$

$\varphi(-\alpha) = -\varphi(\alpha) = 0$

$\Rightarrow -\alpha \in \ker\varphi$

So <u>$\ker\varphi \subseteq R$ is an ideal.</u>

CONSTRUCTION

Let $I \subseteq R$ be

$\Rightarrow (I,+) \leq (R,+)$

$\Rightarrow (I,+) \trianglelefteq (R,+)$

$+$ is comm.

$(R/I,+)$ is a group

We can define a ring structure on $(R/_I, +)$ as follows

$$(r+I)(s+I) = rs+I$$

Is this well-defined?

Suppose that $r'+I = r+I$ ①
and
$s'+I = s+I$ ②

Is it true that
$rs+I = r's'+I$

$r'+I = r+I$
$\Leftrightarrow r'-r \in I$
$\exists \varepsilon \in I$

$r'-r = \varepsilon$

$s'+I = s+I$
$\Leftrightarrow s'-s \in I$
$\Leftrightarrow \exists \delta \in I$ s.t.
$s'-s = \delta$
$\Leftrightarrow s' = s+\delta$

- $r's'+I = (r+\varepsilon)(s+\delta)+I$
$= (rs + \underbrace{\varepsilon s + r\delta + \varepsilon\delta}_{\in I}) + I$
$= rs+I$

So we have an addition and multiplication on $R/_I$

<u>CLAIM</u> $(R/_I, +, \cdot)$ is a ring

$[(r+I)(s+I)][t+I] = (rs+I)(t+I)$
$= (rs)t + I = r(st)+I$
$= (r+I)+(s+I) = (r+I)((s+I)(t+I))$ ASSOC.

<u>DISTR. 1</u>

$(r+I)((a+I)+(b+I))$
$= (r+I)((a+b)+I)$
$= r(a+b)+I$
$= (r.a+r.b)+I = (r.a+I)+(r.b+I)$
$= (r+I)(a+I)+(r+I)(b+I)$

<u>DISTR. 2</u>

$((a+I)+(b+I))(r+I)$
$= (a+I)(r+I)+(b+I)(r+I)$

- If R has a unity then $R/_I$ has a unity.
(If $1.r = r.1 = r$ $\forall r \in R$ then $(1+I)(r+I) = (r+I)(1+I) = r+I$)

- Similarly if R is comm then so is $R/_I$
$(r+I)(s+I) = rs+I = sr+I = (s+I)(r+I)$

Moreover there is a canonical homom.

$$\pi : R \twoheadrightarrow R/_I$$
$$r \longrightarrow r+I$$

- $\pi(r+s) = (r+s)+I$
$= (r+I)+(s+I)$
$= \pi(r)+\pi(s)$
- $\pi(r.s) = rs+I = (r+I)(s+I)$
$= \pi(r).\pi(s)$
- $\pi(1) = 1+I$

$\ker \pi = \{\alpha \in R \mid \alpha+I = 0 \text{ in } R/_I\} = I$

$$\begin{tikzcd} R \arrow[r, two heads, "\pi"] \arrow[dr, "\varphi"'] & R/I \arrow[d, dashed, "\psi"] \\ & S \end{tikzcd}$$

$\exists\, \psi : R/I \longrightarrow S$

s.t. $\psi \circ \pi = \varphi$

$\Leftrightarrow I \subseteq \ker \varphi$

Suppose

$\exists\, \psi$ s.t

$\psi \circ \pi = \varphi$

if $\alpha \in I \Rightarrow$
$\alpha \in \ker \pi$

$$\begin{aligned} \varphi(\alpha) &= (\psi \circ \pi)(\alpha) \\ &= \psi(\pi(\alpha)) \\ &= \psi(0) \\ &= 0 \end{aligned}$$

$\Longrightarrow \alpha \in \ker \varphi$

$\therefore I \subseteq \ker \varphi$

Suppose that $I \subseteq \ker \varphi$

Define $\psi : R/_I \to S$

as $\psi(r+I) = \varphi(r)$

Is this well-defined?

If $r' + I = r + I$ then $\exists\, \varepsilon \in I$ s.t

$r' = r + \varepsilon$

$$\begin{aligned} \psi(r'+I) = \varphi(r') = \varphi(r+\varepsilon) &= \varphi(r) + \varphi(\varepsilon) \\ &= \varphi(r) + 0 = \varphi(r) = \psi(r+I) \end{aligned}$$

$\Rightarrow \varepsilon \in I \subseteq \ker \varphi$

$\therefore$ So ψ is well-defined.

ψ is a homomorphism

▽ $\psi((r+I)+(s+I))$

$$\begin{aligned} &= \psi((r+s)+I) \\ &= \varphi(r+s) = \varphi(r) + \varphi(s) \\ &= \psi(r+I) + \psi(s+I) \end{aligned}$$

▽ $\psi((r+I)(s+I))$

$$\begin{aligned} &= \psi(rs+I) = \varphi(rs) \\ &= \varphi(r).\varphi(s) \\ &= \psi(r+I)\,\psi(s+I) \end{aligned}$$

▽ $\psi(1+I) = \varphi(1) = 1$

Note if such a ψ exists then it is unique since π is surjective

▶ What are the ideals of $\mathbb{Z}$?

$n\mathbb{Z}$, for some $n \in \mathbb{N}_0$

▶ What are the ideals of F, if F is a field

if $(0) \subsetneq I \subseteq F$

then $\exists\, \alpha \in I \setminus \{0\}$

$\Longrightarrow 1 = \underset{\in F}{\alpha^{-1}} \cdot \underset{\in I}{\alpha} \in I$

If $r \in F$, $r = \underset{\in F}{r} \cdot \underset{\in I}{1} \in I$

$\Rightarrow I = F$

$\mathbb{Z} \subseteq \mathbb{Q}$

Field of Fractions

Let A be an integral domain will construct a field F s.t

$i : A \hookrightarrow F$

and it will have the property

$$\begin{tikzcd} A \arrow[r, hook, "i"] \arrow[dr, "J"'] & F \arrow[d, dashed, "\varphi"] \\ & K \text{ -field} \end{tikzcd}$$

CONSTRUCTION

$F = \left\{ \frac{a}{b} \;\middle|\; a \in A, b \in A \setminus \{0\} \right\}$

$\frac{a}{b} \sim \frac{c}{d} \Leftrightarrow ad - bc = 0$

$\frac{a}{b} + \frac{a'}{b'} = \frac{ab' + ba'}{bb'}$

$\frac{a}{b} \cdot \frac{a'}{b'} = \frac{a \cdot a'}{b \cdot b'}$

These are well-defined

▷ If $\frac{a}{b} \sim \frac{c}{d}$ $\frac{a'}{b'} \sim \frac{c'}{d'}$

$ad = bc$

$a'd' = b'c'$

$\frac{a}{b} + \frac{a'}{b'} = \frac{ab' + a'b}{bb'}$

$\frac{c}{d} + \frac{c'}{d'} = \frac{cd' + c'd}{dd'}$

$$\frac{ab'+a'b}{bb'} \sim \frac{cd'+c'd}{dd'} \iff (ab'+a'b)\,dd' \overset{?}{=} (cd'+c'd)\,bb'$$

$$\iff adb'd' + a'd'bd \overset{?}{=} bcd'b' + b'c'bd$$

$$\| \quad bcb'd' + b'c'bd$$

$$\frac{0}{a} = \frac{0}{1} = 0$$

if $\frac{a}{b} \in F \setminus \{0\}$

$\Longrightarrow a = 0$

$\frac{b}{a} \in F \quad \frac{b}{a} \cdot \frac{a}{b} = \frac{1}{1}$ unit in F

F is a field

$$i : A \hookrightarrow F$$
$$a \longrightarrow \frac{a}{1}$$

$$\ker i = \{ a \in A \mid i(a) = \tfrac{0}{1} \text{ in } F \}$$
$$= \{ a \in A \mid \tfrac{a}{1} = \tfrac{0}{1} \text{ in } F \}$$
$$= \{0\}$$

Now suppose

$$\begin{tikzcd} A \arrow[r, hook, "i"] \arrow[dr, "J"'] & F \arrow[d, dashed, "\varphi"] \\ & K \end{tikzcd}$$

field of fractions

field

UNIQUENESS

$$\varphi\left(\frac{a}{b}\right) = \varphi\left(\frac{a}{1}\frac{1}{b}\right)$$
$$= \varphi\left(\frac{a}{1}\right)\varphi\left(\frac{1}{b}\right)$$
$$= \underbrace{\varphi \circ i(a)} \cdot \varphi(i(b))^{-1}$$
$$= J(a)(J(b))^{-1}$$

DEFINE

$$\varphi\left(\frac{a}{b}\right) = J(a) \cdot J(b)^{-1}$$

is this well-defined?

$$\frac{a}{b} = \frac{c}{d} \Rightarrow ad = bc$$

$b \neq 0$
$d \neq 0$

$$\Rightarrow J(a) \cdot J(d) = J(b)\, J(c)$$

$$\Rightarrow J(b) \neq 0,\ J(d) \neq 0$$

J is injective

$$J(a)\, J(b)^{-1} = J(c)\, J(d)^{-1}$$
$$\| \qquad\qquad \|$$
$$\varphi\left(\frac{a}{b}\right) \qquad \varphi\left(\frac{c}{d}\right)$$

In this lecture suppose that R is a commutative ring with unity.

Let $R[x] = \{ a_0 + a_1 x + \dots + a_n x^n \mid a_0, \dots, a_n \in R \}$

For example, if $R = \mathbb{Z}$

then $2 - 3x \in \mathbb{Z}[x]$

$-2 + 5x^3 - 7x^8 \in \mathbb{Z}[x]$

We define addition in $R[x]$ as follows

$$(a_0 + a_1 x + \dots + a_n x^n) + (b_0 + b_1 x + \dots + b_m x^m)$$
$$= (a_0 + b_0) + (a_1 + b_1)x + \dots + (a_\ell + b_\ell) x^\ell$$

s.t. $\ell = \max(n, m)$

if $n < r$, $a_r = 0$

$m < r$, $b_r = 0$

$$(2 - 3x) + (-2 + 5x^3 - 7x^8) = -3x + 5x^3 - 7x^8$$

$$\left[R[x] = \{ \sum_{0 \le i} a_i x^i \mid a_i = 0 \text{ except for finitely many} \} \right]$$

$$\left[\sum_{0 \le i} a_i x^i + \sum_{0 \le i} b_i x^i = \sum_{0 \le i} (a_i + b_i) x^i \in R[x] \right]$$

We can also define an multiplication on $R[x]$, as

$$\sum_{0 \le i} a_i x^i \sum_{0 \le j} b_j x^j = \sum_{0 \le n} c_n x^n$$

$$c_n = \sum_{0 \le i \le n} a_i b_{n-i}$$

$$(2 - 3x)(-2 + 5x^3 - 7x^8)$$
$$= -4 + 6x + 10x^2 + \dots$$

PROPOSITION

$R[x]$ is a commutative ring with unity.

PROOF

$0 = 0 + 0 \cdot x + \dots$

$$\left(\sum_{0 \le i} \alpha_i x^i \sum_{0 \le i} \beta_i x^i \right) \left(\sum_{0 \le i} \gamma_i x^i \right) = \sum_{0 \le j} \left(\sum_{0 \le i \le j} \alpha_i \beta_{j-i} \right) x^j \sum_{0 \le i} \gamma_i x^i = \sum_{0 \le k} \sum_{0 \le i \le j} (\alpha_i \beta_{j-i}) \gamma_{k-j} x^k$$

$$\sum_{0 \le j \le k} \left(\sum_{0 \le i \le j} \alpha_i \beta_{j-i} \right) \gamma_{k-j} = \sum_{0 \le i_1, i_2, i_3} (\alpha_{i_1} \beta_{i_2}) \gamma_{i_3}$$

$$\left(\sum_{0 \le i} \alpha_i x^i \right) \left(\sum_{0 \le i} \beta_i x^i \sum_{0 \le i} \gamma_i x^i \right) = \sum_{0 \le i} \alpha_i x^i \sum_{0 \le i} \left(\sum_{b+c=i} \beta_b \gamma_c \right) x^i$$

$$= \sum_{0 \le i} \left(\sum_{\substack{0 \le a, b, c \\ a+b+c=i}} \alpha_a (\beta_b \gamma_c) \right) x^i$$

Define

$$\deg : R[x] \to \mathbb{N}_{\geq 0} \sqcup \{-\infty\}$$

- $\deg(0) = -\infty$

$$\deg\left(\sum_{0 \le i} a_i x^i \right) = \max \{ i \mid a_i \neq 0 \}$$

$$\deg(f + g) \le \max(\deg(f), \deg(g))$$

- If R is an integral domain then

$$\deg(f(x) \cdot g(x)) = \deg((f(x)) + \deg((g(x))$$

Warning

This need not be true if R is not an integral domain

$R = \mathbb{Z}_6$,

$2x, 3x \in \mathbb{Z}_6[x]$

$(2x)(3x) = 6x^2 = 0$

$\deg(2x \cdot 3x) = -\infty$

Suppose R is an integral domain

$$\underbrace{(a_0 + \dots + a_n x^n)}_{f(x)}\underbrace{(b_0 + \dots + b_m x^m)}_{g(x)} = a_0 b_0 + \dots + (a_n b_m) x^{n+m}$$

$\deg f(x) = n$, $a_n \neq 0$

$\deg g(x) = m$, $b_m \neq 0$

$a_n \cdot b_m \neq 0$, $\neq 0$, $\neq 0$

$\deg f(x) g(x) = n + m = \deg f(x) + \deg g(x)$

Let us look at the case

$F[x]$ — field

In this case we have the division algorithm

PROPOSITION

Let $a(x), b(x) \in F[x]$, $b(x) \neq 0$

then there exist unique $q(x)$ and $r(x)$ s.t.

$a(x) = b(x) q(x) + r(x)$ s.t.

$\deg r(x) < \deg b(x)$

$a, b \in \mathbb{Z}$, $b \neq 0$

$\exists!\ q, r \in \mathbb{Z}$ s.t. $0 \leq r < |b|$

and

$a = bq + r$

Why are $q(x)$ and $r(x)$ unique?

Suppose that $\tilde{q}(x)$ and $\tilde{r}(x)$ also satisfy

$a(x) = b(x)\tilde{q}(x) + \tilde{r}(x)$ with $\deg \tilde{r}(x) < \deg b(x)$

$b(x)q(x) + r(x) = a(x) = b(x)\tilde{q}(x) + \tilde{r}(x)$

$$b(x)(q(x) - \tilde{q}(x)) = \underbrace{\tilde{r}(x) - r(x)}_{< \deg b(x)}$$

$\Longrightarrow q(x) - \tilde{q}(x) = 0$

$\Longrightarrow q(x) = \tilde{q}(x)$

and

$r(x) = \tilde{r}(x)$

▶ $\mathbb{Z}[x]$ $\quad x^2 + 3 \mid 2x + 1$

It is important to have a field here because it does not contain inverse of 2

DEF: Let $a(x), b(x) \in F(x)$

We say that $b(x) \mid a(x)$ (divides)

if $a(x) = b(x) q(x)$

PROPOSITION

If $I \subseteq F[x]$ is an ideal then $I = (f(x))$
$= \{ f(x) g(x) \mid g(x) \in F[x] \}$

PROOF

Trivial if $I = \{0\}$

Suppose $I \neq 0$

If $I \subseteq \mathbb{Z}$ an ideal then $I = n\mathbb{Z}$ for some $n \in \mathbb{Z}$
$= \{ na \mid a \in \mathbb{Z} \}$

Let $\min \{ \deg a(x) \mid a(x) \in I \setminus \{0\} \} = m$

Let $f(x) \in I \setminus \{0\}$ s.t. $\deg f(x) = m$

then if $\alpha(x) \in I$

then $\alpha(x) = f(x) q(x) + r(x)$ s.t. $\deg r(x) < \deg f(x) = m$

$\Rightarrow r(x) \in I$, $\deg r(x) < m \Rightarrow r(x) = 0$

$\alpha(x) = f(x)q(x)$

$\Rightarrow \alpha(x) \in (f(x))$

$\therefore I \subseteq (f(x))$

$\Rightarrow I = (f(x))$

$(f(x)) \subseteq I$

Observation

Let $\alpha \in F$, we have $\varphi_\alpha : F[x] \to F$

$f(x) \to f(\alpha)$

$\varphi_\alpha(f(x)) = f(\alpha)$

$\varphi_\alpha\left(\sum_{0 \le i} a_i x^i\right) = \sum_{0 \le i} a_i \alpha^i \in F$

$\ker \varphi_\alpha = \{f(x) \mid f(\alpha) = 0\}$

$f(x) = (x - \alpha)q(x) + \lambda$

$\lambda \in F$

$f(\alpha) = (\alpha - \alpha)q(\alpha) + \lambda = \lambda$

$\Rightarrow f(x) = (x - \alpha)q(x) + f(\alpha)$

So $f(\alpha) = 0 \Leftrightarrow (x - \alpha) \mid f(x)$

$\Leftrightarrow (x - \alpha) \in f(x)$

OBSERVATION

$(F[x])^\times = F^\times$ invertible elements

DEF: Let $f(x) \in F[x] \setminus F$ (i.e $\deg f \geq 1$) then we say that

$f(x)$ is <u>irreducible</u> if whenever $f(x) = a(x)b(x)$ then

$a(x) \in F[x]^\times$ OR $b(x) \in F[x]^\times$

▶ (1) If $\deg f(x) = 1$ then $f(x)$ is irreducible

(2) $x^2 - 3x + 2 \in \mathbb{Q}[x]$

$= (x-1)(x-2)$ is <u>not</u> irreducible.

214 **THEOREM** If $f(x) \in F[x]$ and $2 \le \deg f(x) \le 3$ then f is irreducible $\Longleftrightarrow$ f does not have a root in F, i.e.

$f(a) \neq 0$, $\forall a \in F$

PROOF

($\Rightarrow$) If $f(x)$ is irreducible

Let $a \in F$

$f(x) = (x - a)q(x) + f(a)$

if $f(a) = 0 \Rightarrow f(x) = (x - a)q(x)$

($\Leftarrow$) Suppose $\forall a \in F$ $f(a) \neq 0$

Suppose $f(x)$ is not irreducible then $f(x) = a(x)b(x)$,

w/ $1 \le \deg a(x)$

$1 \le \deg b(x)$

WLOG $\deg a(x) = 1$

$a(x) = ax + b \Rightarrow f(-b/a) = 0$

▶ $(x^2+1)^2 \in \mathbb{R}[x]$ does not have a root in $\mathbb{R}$ but not irreducible.

Recall

- We fixed a field F. We want to understand $F[x]$
- Note that $(F[x])^\times = F^\times$

- We have showed that if $I \subseteq F[x]$ then $\exists f(x) \in F[x]$ s.t. $I = (f(x))$

DEF

Suppose that $f(x), g(x) \in F[x] \setminus \{0\}$

We say that $d(x)$ is <u>a</u> greatest common divisor if

$d(x) \mid f(x)$ & $d(x) \mid g(x)$ also has the property

that whenever $e(x) \mid f(x)$ and $e(x) \mid g(x) \Rightarrow e(x) \mid d(x)$

$\Longleftrightarrow \{e(x) \mid e(x) \mid f(x) \text{ and } e(x) \mid g(x)\} = \{e(x) \mid e(x) \mid d(x)\}$

Does a gcd exist and is it unique?

<u>Observation</u>

Even if a gcd of $f(x)$ and $g(x)$ exists, it is $\boxed{\text{not unique}}$

b/c if $d(x)$ satisfy the above hypothesis then

so does $\lambda . d(\lambda)$ for any $\lambda \in F[x]^{\times} = F^{\times}$

(Note that if $\lambda \in F[x]^{\times}$ then $a(x) \mid b(x) \Leftrightarrow \lambda\, a(x) \mid b(x)$
$\Leftrightarrow a(x) \mid \lambda\, b(x)$

<u>Lemma</u>

If $d(x)$ and $\tilde{d}(x)$ are two greatest common divisors of $f(x)$ and $g(x)$ then $\exists\, \lambda \in F[x]^{\times}$ s.t. $\tilde{d}(x) = \lambda d(x)$

<u>PROOF</u>

Since $\tilde{d}(x)$ is a common divisor of $f(x)$ and $g(x)$

$\Rightarrow \tilde{d}(x) \mid f(x)$ and $\tilde{d}(x) \mid g(x)$

$\Rightarrow \tilde{d}(x) \mid d(x) \qquad (1)$

$d(x)$ is a gcd.

By symmetry

$d(x) \mid \tilde{d}(x) \qquad (2)$

$(1) \Rightarrow d(x) = \tilde{d}(x) . a(x)$

$(2) \Rightarrow \tilde{d}(x) = d(x) . b(x)$

$d(x) = a(x) b(x) d(x)$

$\Longrightarrow d(x)(1 - a(x) b(x)) = 0$

$\underset{\substack{d(x) \neq 0 \\ F[x] \text{ int. dom}}}{\Longrightarrow} 1 - a(x) b(x) = 0 \Rightarrow a(x) b(x) \in F[x]^{\times}$

<u>Existence of G.C.D</u>

Let $\underset{\neq 0}{f(x)}, \underset{\neq 0}{g(x)}$

then $I = (f(x), g(x))$

$= \{\underline{a(x) f(x)} + \underline{b(x) g(x)} \mid a(x), b(x) \in F[x]\}$

$\subseteq F[x]$

$\underset{\substack{\text{ideals in } F[x] \\ \text{are principal} \\ \text{generated} \\ \text{by an element}}}{\Longrightarrow} \exists\, d(x)$ s.t

$(d(x)) = (f(x), g(x)) = I$

<u>CLAIM</u> $d(x)$ is a gcd of $f(x)$ and $g(x)$.

<u>PROOF</u> $f(x), g(x) \in I = (d(\lambda)) \Rightarrow d(x) \mid f(x)$ and $g(x)$

Suppose $e(x) \mid f(x)$ and $e(x) \mid g(x)$

$\Rightarrow \exists\ r(x), s(x) \in F[x]$ s.t.

$$e(x) \cdot r(x) = f(x)$$
$$e(x) \cdot s(x) = g(x)$$

But $d(x) \in (f(x), g(x))$

$\Rightarrow \exists\ a(x), b(x)$ s.t.

$$\Rightarrow d(x) = a(x) f(x) + b(x) g(x)$$
$$= a(x) r(x) e(x) + b(x) s(x) e(x)$$
$$= (a(x) r(x) + b(x) s(x)) e(x)$$
$$\Rightarrow e(x) \mid d(x)$$

Therefore gcd of $f(x)$ and $g(x)$ exist

Moreover, if $d(x)$ is a gcd then

$$(f(x), g(x)) = (d(x))$$

DEF We say that $f(x)$ and $g(x)$ are relatively prime if $\gcd(f(x), g(x)) = 1$

$$\Longleftrightarrow (f(x), g(x)) = F[x]$$
$$\Longleftrightarrow \exists\ a(x), b(x) \in F[x] \text{ s.t. } a(x) f(x) + b(x) g(x) = 1$$

Lemma

Suppose that $p(x) \in F[x]$ is irreducible and $p(x) \mid f(x) g(x)$ then either $p(x) \mid f(x)$ or $p(x) \mid g(x)$

PROOF

Suppose $p(x) \nmid f(x)$

$$d(x) = (p(x), f(x))$$
$$d(x) \mid p(x) \Rightarrow$$

$\exists\ a(x)$ s.t.

$$d(x) a(x) = p(x)$$

$\Rightarrow$ either $d(x) \in F[x]^\times$

or $a(x) \in F[x]^\times$

Suppose $a(x) \in F[x]^\times$

$$\Rightarrow p(x) \mid d(x),\ d(x) \mid f(x)$$
$$\Rightarrow p(x) \mid f(x) \quad \text{since } d(x) = (p(x), f(x))$$

$\therefore\ d(x) \in F[x]^\times \Rightarrow 1 = (p(x), f(x))$

$\Longrightarrow \exists\ a(x), b(x) \in F[x]$ s.t. $1 = a(x) p(x) + b(x) f(x)$

$$\Longrightarrow g(x) = \underbrace{g(x) a(x) \cdot p(x)}_{p(x) \mid} + \underbrace{b(x) f(x) g(x)}_{p(x) \mid} \Rightarrow p(x) \mid g(x)$$

$\boxed{2/7}$ **THEOREM** (Analog of the main theorem of arithmetic)

Let $f(x) \in F[x] \setminus F$

then $f(x) = p_1(x) \dots p_r(x)$ s.t. all $p_i(x)$ are irreducible

Moreover <u>this decomposition into irreducibles is unique up to ordering and up to multiplication by a unit.</u>

PROOF

Existence

Suppose that there is an $f(x) \in F(x) \setminus F$ s.t. $f(x)$ cannot be written as a product of irreducibles.

$$\mathcal{S} = \left\{ a(x) \;\middle|\; \begin{array}{l} a(x) \in F[x] \setminus F \\ a(x) \text{ cannot be written} \\ \text{as a product of irreducibles} \end{array} \right\}$$
$$\neq 0$$

Let $r(x) \in \mathcal{S}$ s.t. $\deg r(x)$ is smallest among all $a(x) \in \mathcal{S}$

$r(x)$ is not irreducible then

$$r(x) = s_1(x)\, s_2(x) \quad \text{s.t. } \deg s_i(x) \geq 1$$

$$\Longrightarrow \deg s_i(x) < \deg r(x)$$

$$\Longrightarrow s_i(x) \notin S$$

$$\Longrightarrow s_1(x) = t_1(x) \cdots t_k(x)$$

$$s_2(x) = q_1(x) \cdots q_\ell(x)$$

s.t. $t_i(x), q_j(x)$ are irred.

$$r(x) = s_1(x) s_2(x) = t_1(x) \cdots t_k(x)\, q_1(x) \cdots q_j(x)$$

UNIQUENESS

Suppose

① $f(x) = p_1(x) \cdots p_r(x)$

② $f(x) = q_1(x) \cdots q_s(x)$

① $\Rightarrow p_1(x) \mid f(x) = q_1(x) \cdots q_s(x)$

$\Rightarrow p_1(x) \mid q_i(x)$ WLOG assume $i=1$

A poly. previous lemma (s-1) times

$\Longrightarrow p_1(x) \mid q_1(x) \Longrightarrow \exists\, \lambda \in F[x]^\times$ s.t.

both irred $\quad q_1(x) = \lambda\, p_1(\lambda)$

cancel $p_1(x)$, continue